Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 270
Manufacturing Technology

Lecture 3: Metal Casting

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Course Objectives

- Understand the fundamentals of metal casting
- Calculate solidification time
- Calculate melting flow rate
- Describe various metal casting processes

✓ Die casting

✓ Investment casting

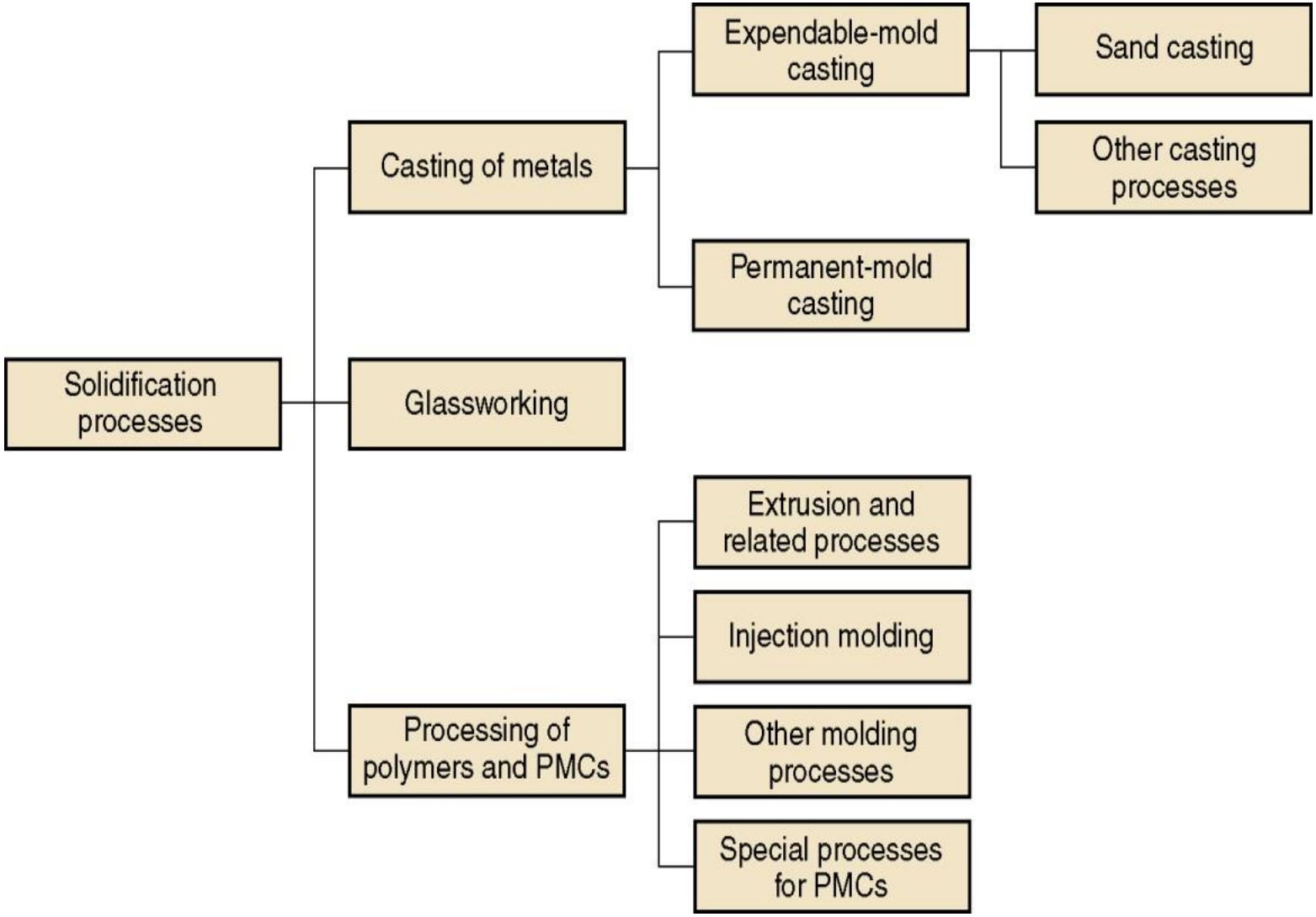
✓ Sand casting

Solidification Process

- Starting work material for casting is either a liquid or is in a highly plastic condition, and a part is created through solidification of the material
- Solidification processes can be classified according to engineering material processed:
  - ✓Metals
  - ✓Ceramics, specifically glasses
  - ✓Polymers and polymer matrix composites (PMCs)

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Classification of solidification processes



Casting of metals

- Casting is the process in which molten metal flows by gravity or other force into a mold where it solidifies in the shape of the mold cavity
- The term ***casting*** also applies to the part made in the process
- Steps in casting can be as simple as:
    1. Melt the metal
    2. Pour it into a mold
    3. Let it freeze

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Advantages of Casting

- It can create complex part geometries
- It can create both external and internal shapes
- Some casting processes are ***net shape***; others are ***near net shape***
- It can produce very large parts
- Some casting methods are suited to mass production

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Disadvantages of Casting

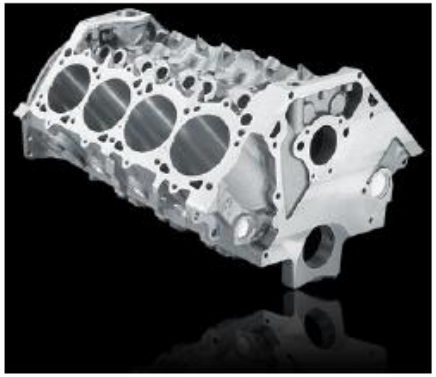
- Different disadvantages occur for different casting processes.
- Generally, disadvantages of casting include:
  - ✓ Limitations on mechanical properties
  - ✓ Poor dimensional accuracy and surface finish for some processes; e.g., sand casting
  - ✓ Safety hazards to workers due to hot molten metals
  - ✓ Environmental problems

Parts Made by Casting

- **Big parts** such as Engine blocks and heads for automotive vehicles, wood burning stoves, machine frames, railway wheels, pipes, church bells, big statues, pump housings
- **Small parts** such as Dental crowns, jewelry, small statues, frying pans









Fundamentals of Casting Technology

- Casting is usually performed in a **foundry**, which is a factory equipped for
  - ✓ Making molds
  - ✓ Melting and handling molten metal
  - ✓ Performing the casting process
  - ✓ Cleaning the finished casting
- Workers who perform casting are called ***foundrymen***

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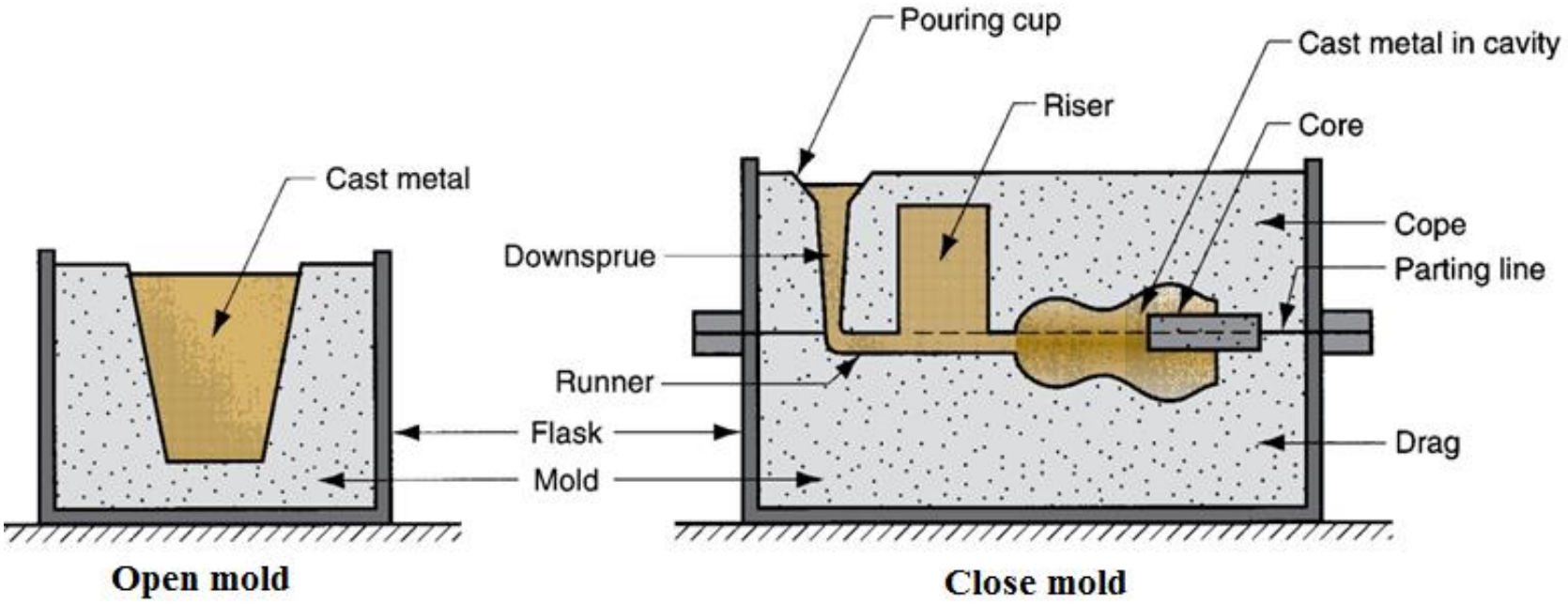
The Mold in Casting

- The mold contains cavity whose geometry determines part shape.
- Actual size and shape of cavity must be slightly enlarged to allow for shrinkage of metal during solidification and cooling.
- Molds are made of a variety of materials, including sand, plaster, ceramic, and metal.

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Types of Molds

- There are two forms of mold:
  - ✓ open mold and
  - ✓ closed mold for more complex mold geometry with gating system leading into the cavity



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Types of Casting Processes

Expendable mold processes: These processes use an expendable mold which must be destroyed to remove casting. The mold materials used include sand, plaster, and similar materials, plus binders. The most common forms of expendable mold casting are sand casting, plaster casting, investment (lost wax) casting, and evaporative pattern casting.

Permanent mold processes: These processes use a permanent mold which can be used to produce many castings. Mode materials are mostly made of metal (or, less commonly, a ceramic refractory material).

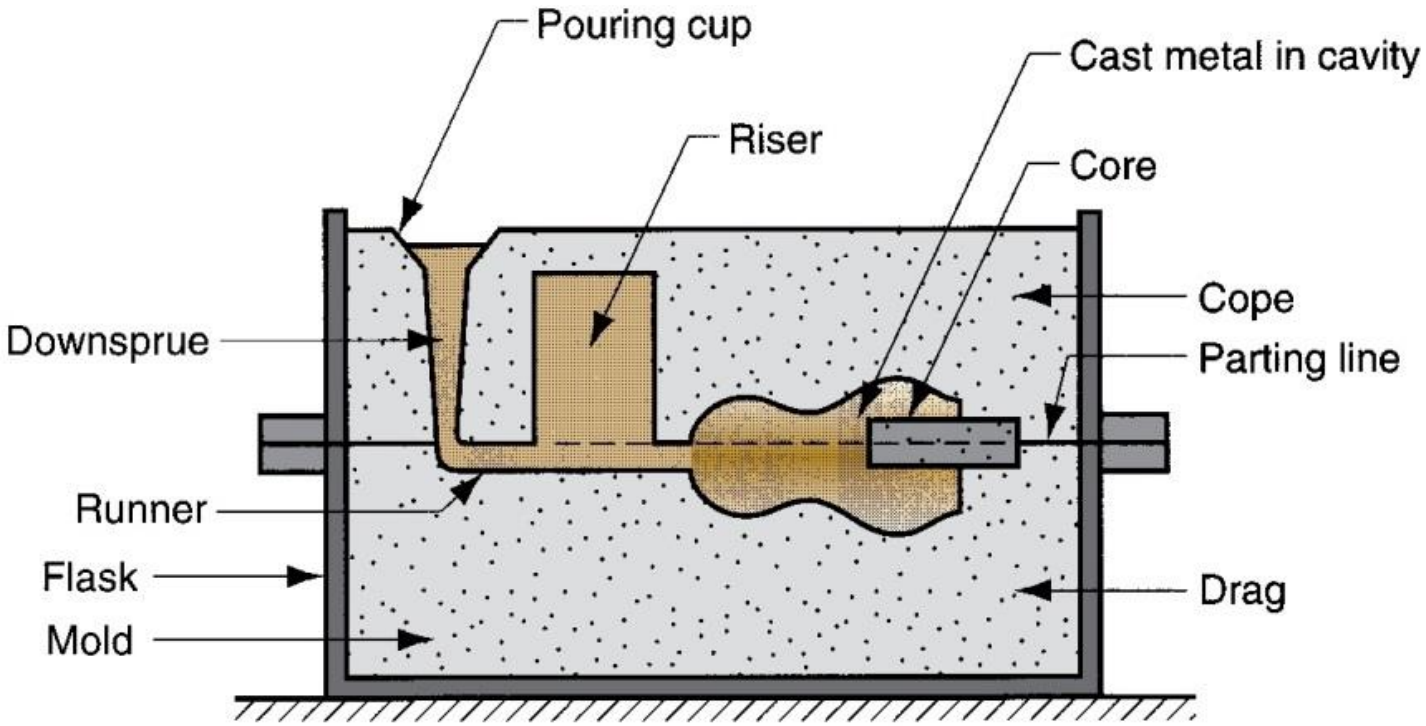
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Advantages and Disadvantages of Casting Processes

- More intricate geometries are possible with expendable mold processes
- Part shapes in permanent mold processes are limited by the need to open the mold
- Permanent mold processes are more economic in high production operations

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Sand Casting Mold



Sand casting mold

- Mold consists of two halves:
  - ✓**Cope**: the upper half of mold
  - ✓**Drag:** the bottom half
- Mold halves are contained in a box, called a ***flask***
- The two halves separate at the ***parting line***

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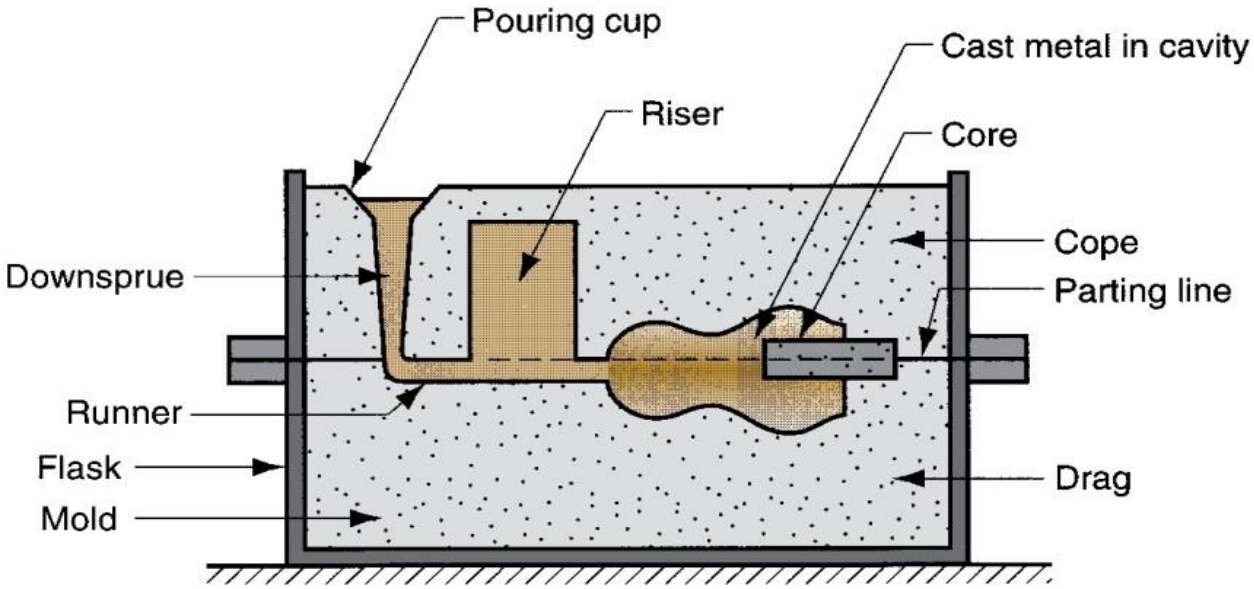
Forming the Mold Cavity in Sand Casting

- Mold cavity is formed by packing sand around a ***pattern***, which has the shape of the part
- When the pattern is removed, the remaining cavity of the packed sand has desired shape of cast part
- The pattern is usually oversized to allow for shrinkage of metal during solidification and cooling
- Sand for the mold is moist and contains a binder to maintain its shape

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Using a Core in the Mold Cavity

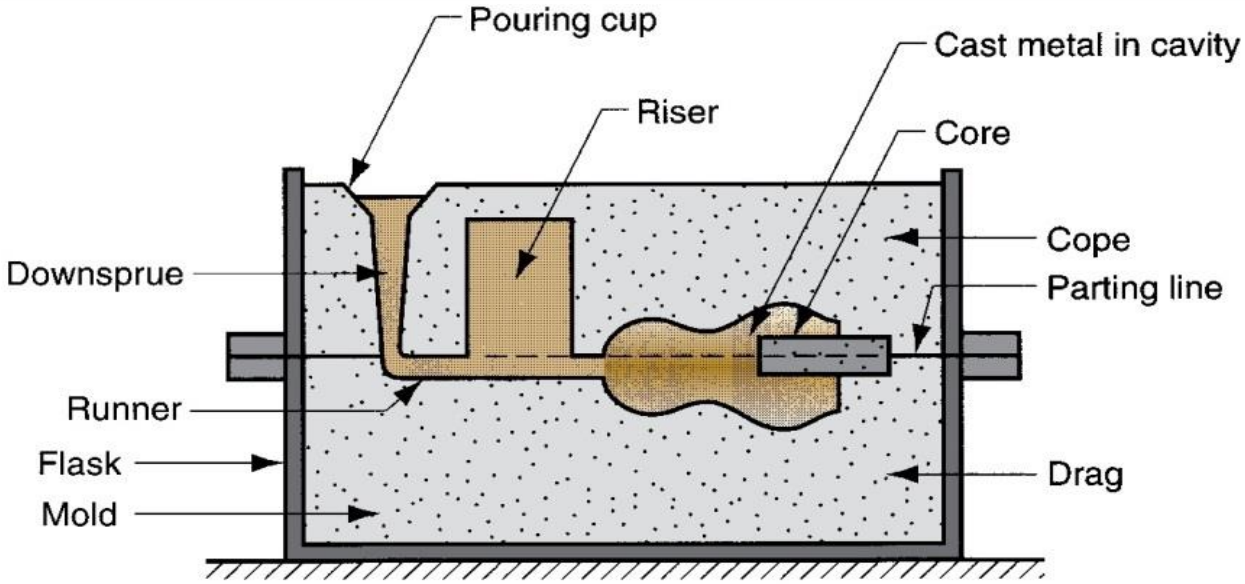
- The mold cavity provides the external surfaces of the cast part
- In addition, a casting may have internal surfaces, determined by a ***core***, placed inside the mold cavity to define the interior geometry of part
- In sand casting, cores are generally made of sand



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Gating System

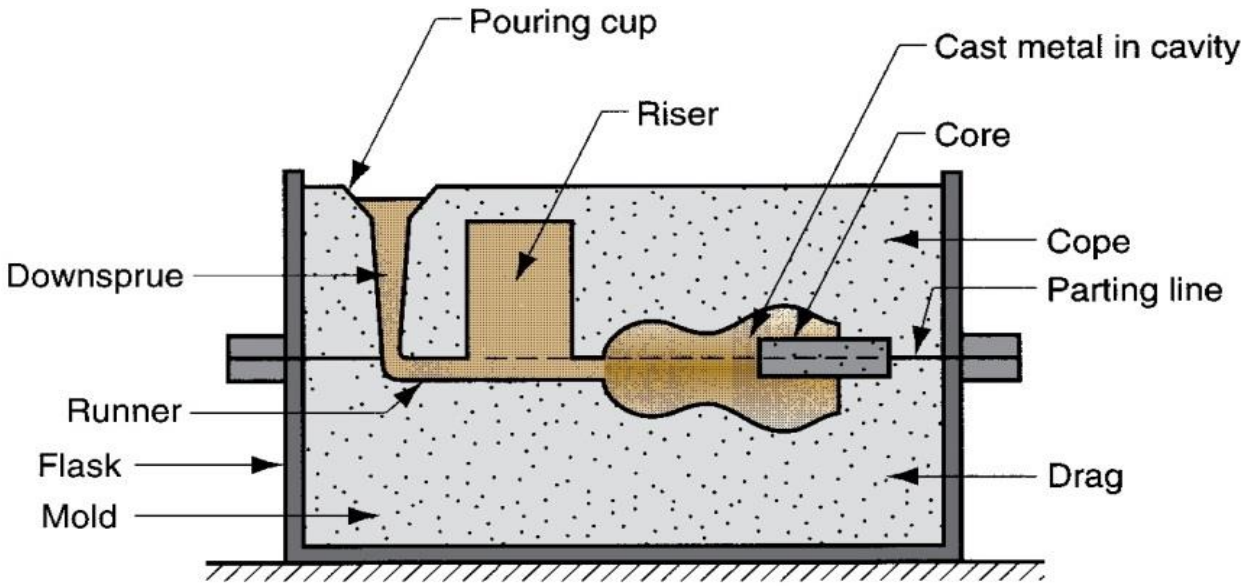
- It is the channel through which molten metal flows into the cavity from the outside of mold
- Consists of a ***downsprue***, through which metal enters a ***runner*** leading to the main cavity
- At the top of the downsprue, a ***pouring cup*** is often used to minimize splash and turbulence as the metal flows into the downsprue



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Riser

- It is a reservoir in the mold which is a source of liquid metal to compensate for shrinkage of the part during solidification
- The riser must be designed to freeze after the main casting in order to satisfy its function



Sand Casting



Heating the Metal

- Heating furnaces are used to heat the metal to molten temperature sufficient for casting.
- The heat required is the sum of:
  1. Heat to raise temperature to melting point
  2. Heat of fusion to convert from solid to liquid
  3. Heat to raise molten metal to desired temperature for pouring.

  This is expressed as

$$H = \rho V\{C_S(T_m - T_a) + H_f + C_l(T_p - T_m)\}$$

(1) (2) (3)

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Heating the Metal

- The heat required can be expressed as

$$H = \rho V\{C_S(T_m - T_a) + H_f + C_l(T_p - T_m)\}$$

(1) (2) (3)

H: total heat required to increase the temperature of the metal to the pouring temp (J)

ρ: density (g/cm^3)

V: Volume of metal used for heating (cm^3)

C_S: Specific heat for the solid (in J/g℃)

T_m : Melting temperature of the metal (℃)

T_a: Ambient temperature (or starting) (℃)

H_f : Heat of fusion (J/g)

C_l: specific heat of the liquid metal (in J/g ℃)

T_p: Temperature of the pouring liquid (℃)

Pouring the Molten Metal

- For this step to be successful, metal must flow into all regions of the mold, most importantly the main cavity, before solidifying
- Factors that determine successful pouring of the molten metal include
  - ✓ Pouring temperature
  - ✓ Pouring rate
  - ✓ Turbulence

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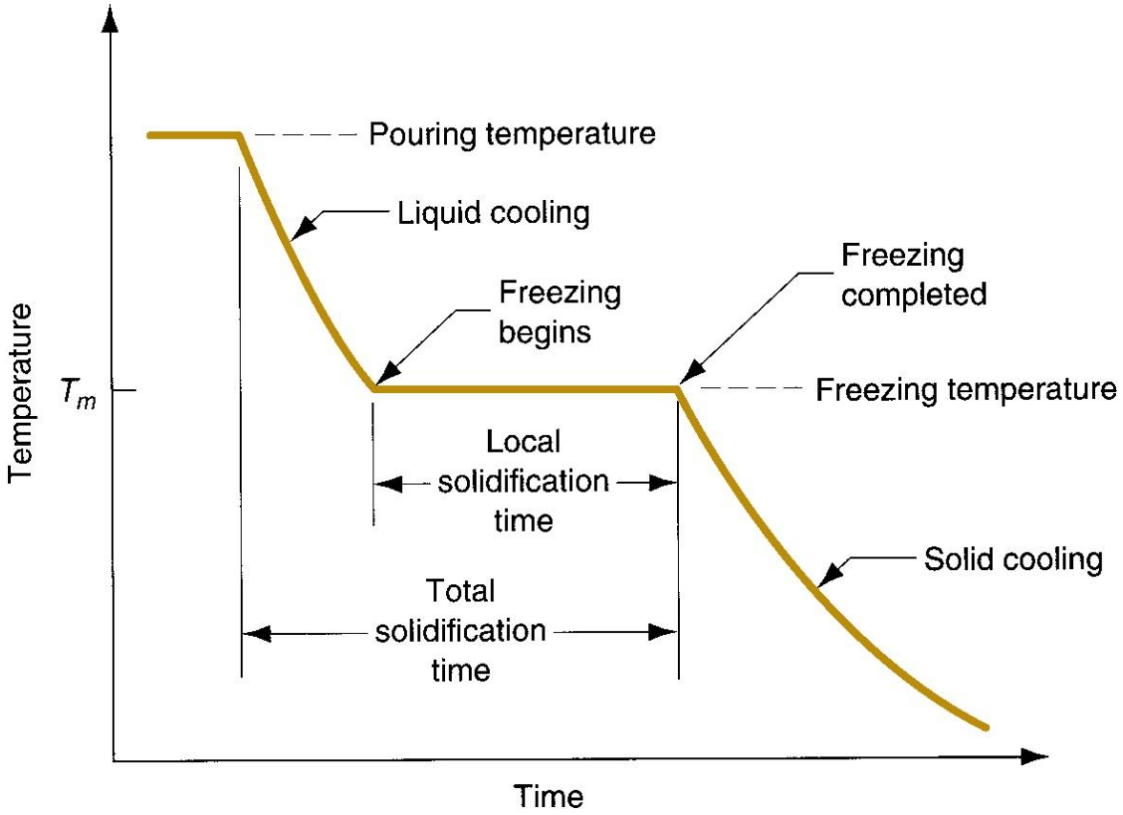
Solidification of metals

- It is the transformation of molten metal back into solid state.
- Solidification differs depending on whether the metal is
  - ✓ A pure element or
  - ✓ An alloy

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Cooling curve for a metal

- A pure metal solidifies at a constant temperature equal to its freezing point (same as melting point)



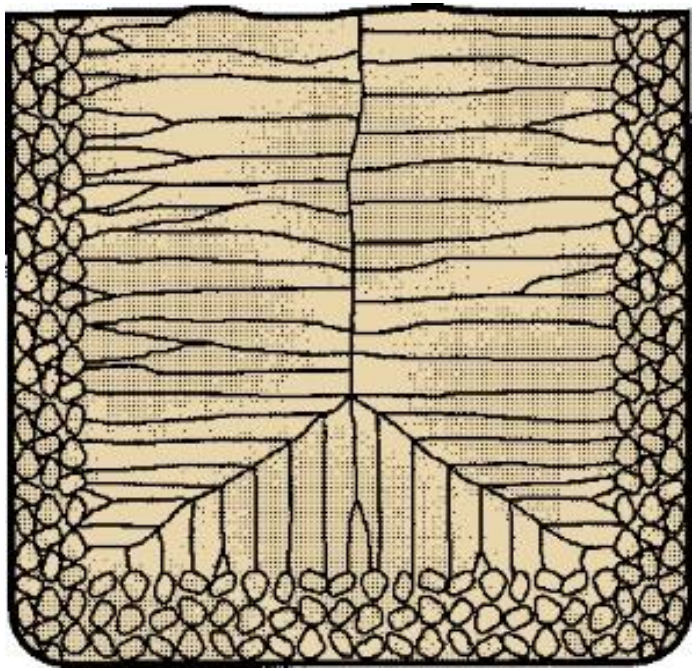
Solidification of Metals

- Due to the chilling action of the mold wall, a thin skin of solid metal is formed at the interface immediately after pouring
- Skin thickness increases to form a shell around the molten metal as solidification progresses
- Rate of freezing depends on heat transfer out of the mold, as well as thermal properties of the metal

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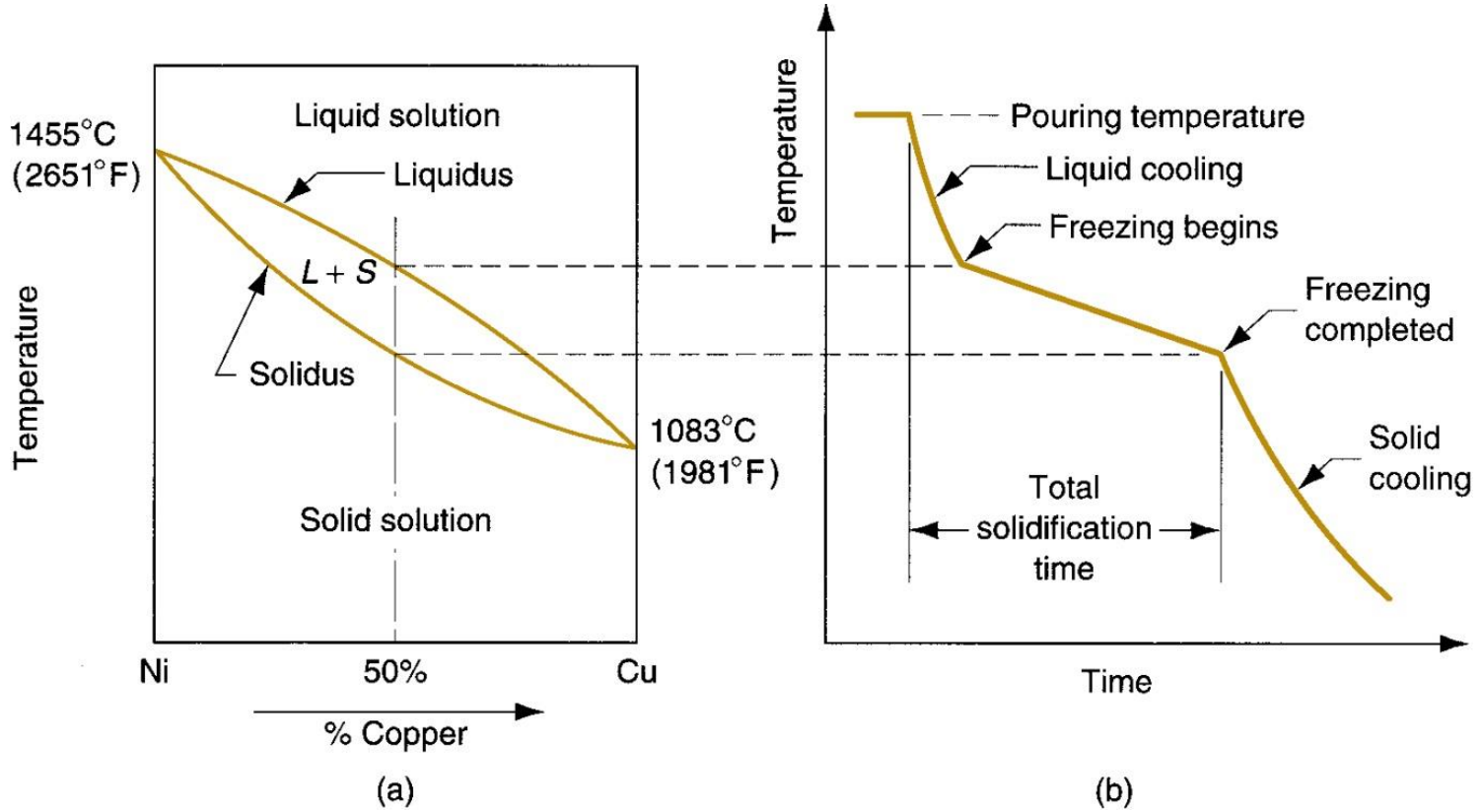
Solidification of metals

- Characteristic grain structure in a casting of a pure metal, showing randomly oriented grains of small size near the mold wall, and large columnar grains oriented toward the center of the casting.



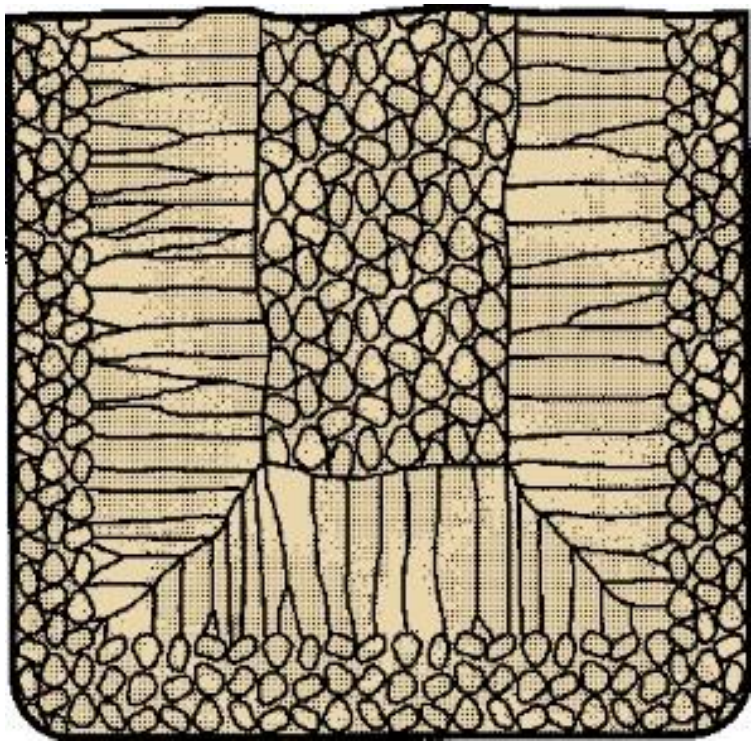
Solidification Temperature Range of Metals

(a) Phase diagram for a copper-nickel alloy system and
(b) cooling curve for a 50%Ni-50%Cu composition



Solidification of Alloys

- Characteristic grain structure in an alloy casting, showing segregation of alloying components in center of casting



Solidification Time

- Total solidification time (T_{TS}) is time required for casting to solidify after pouring.
- T_{TS} depends on size and shape of casting by relationship known as **Chvorinov's Rule**

$$T_{TS} = C_m \left(\frac{V}{A} \right)^n$$

Where T_{TS} is total solidification time

V is volume of the casting

A is surface area of casting

n is exponent with typical value = 2

C_m is ***mold constant***

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Mold Constant in Chvorinov's Rule

- Mold constant C_m depends on:
  - ✓ Mold material
  - ✓ Thermal properties of casting metal
  - ✓ Pouring temperature relative to melting point

- Value of C_m for a given casting operation can be based on experimental data from previous operations carried out using same mold material, metal, and pouring temperature, even though the shape of the part may be quite different.

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Meaning of Chvorinov's Rule

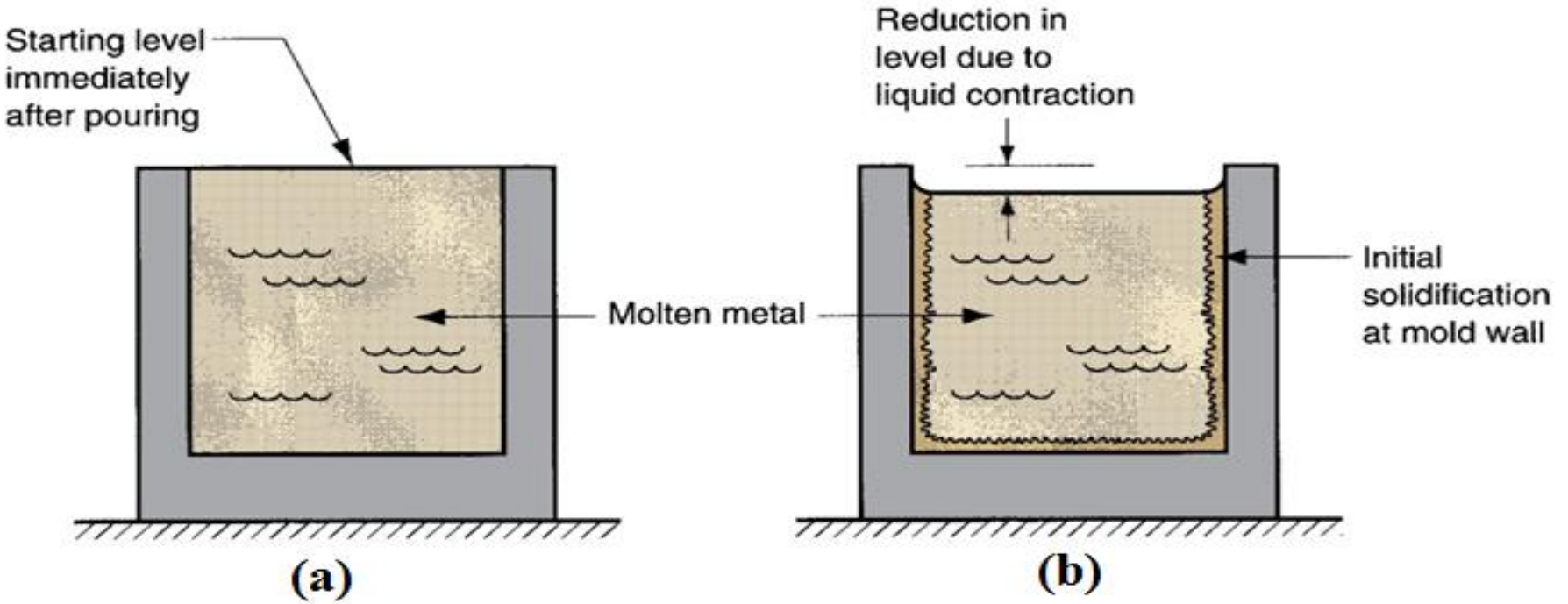
- A casting with a higher volume-to-surface area ratio cools and solidifies more slowly than one with a lower ratio.
- To feed molten metal to the main cavity, T_{TS} for riser must be greater than T_{TS} for main casting.
- Since mold constants of riser and casting will be equal, design the riser to have a larger volume-to-area ratio so that the main casting solidifies first. This minimizes the effects of shrinkage.

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Shrinkage during Solidification and Cooling

(a) Starting level of molten metal after pouring;

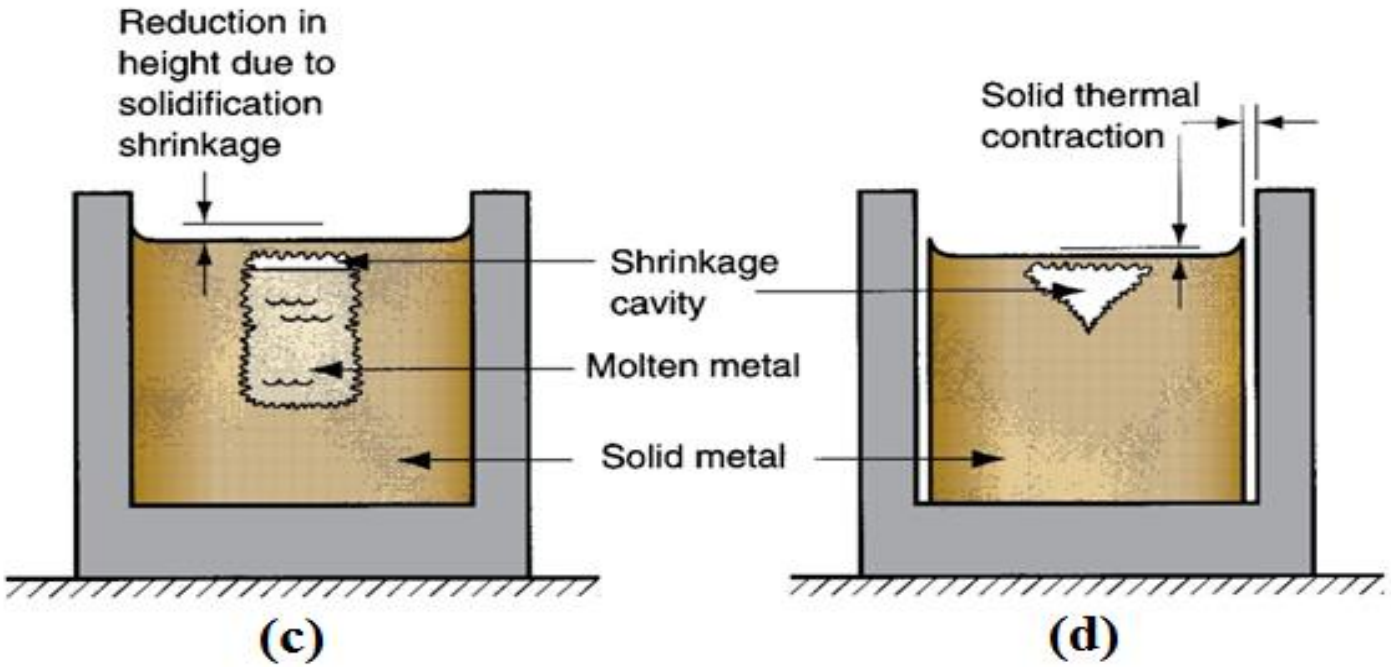
(b) Reduction in level caused by liquid contraction during cooling



Shrinkage during Solidification and Cooling

(c) Reduction in height and formation of shrinkage cavity caused by solidification;

(d) Further reduction in volume due to thermal contraction during cooling of solid metal



Solidification Shrinkage

- Occurs in nearly all metals because the solid phase has a higher density than the liquid phase
- Thus, solidification causes a reduction in volume per unit weight of metal
- Exception: cast iron with high C content
  - ✓ Graphitization during final stages of freezing causes expansion that counteracts volumetric decrease associated with phase change

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Shrinkage Allowance

- Patternmakers correct for solidification shrinkage and thermal contraction by making the mold cavity oversized.
- Amount by which mold is made larger relative to final casting size is called ***pattern shrinkage allowance.***
- Casting dimensions are expressed linearly, so allowances are applied accordingly.

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Directional solidification

- To minimize effects of shrinkage, it is desirable for regions of the casting most distant from the liquid metal supply to freeze first and for solidification to progress from these regions toward the riser(s)
- Thus, molten metal is continually available from risers to prevent shrinkage voids
- The term ***directional solidification*** describes this aspect of freezing and methods by which it is controlled.

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Breakout Room Question

Why is it desirable for regions of the casting most distant from the liquid metal supply to freeze first and for solidification to progress from these regions toward the riser?

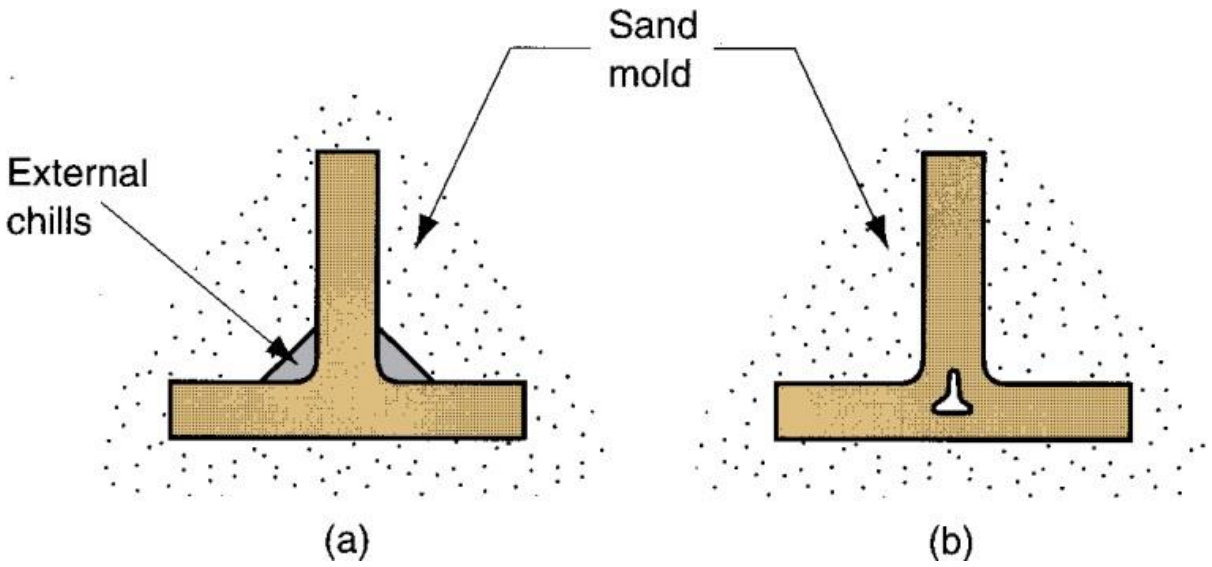
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Achieving Directional Solidification

- Directional solidification is achieved using ***Chvorinov's Rule*** to design the casting, its orientation in the mold, and the riser system that feeds it.
- Locate sections of the casting with lower ***V/A*** ratios away from riser, so freezing occurs first in these regions, and the liquid metal supply for the rest of the casting remains open.

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External Chills



Chills - internal or external heat sinks that cause rapid freezing in certain regions of the casting.

(a) External chill to encourage rapid freezing of the molten metal in a thin section of the casting; and

(b) the likely result if the external chill were not used

Engineering Analysis of Pouring

- There are several relationships that govern the flow of liquid metal through the gating system and into the mold. An important relationship is Bernoulli's theorem, which states that the sum of the energies (head, pressure, kinetic, and friction) at any two points in a flowing liquid are equal.
- Flow velocity: $v = \sqrt{2gh}$;
- Flow rate: $Q = vA$;
- Mold filling time $(T_{MF}) = \frac{V}{Q}$

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Engineering Analysis of Pouring

Flow velocity: $v = \sqrt{2gh}$; (m/s)

$$Q = vA$$

$$T_{MF} = \frac{V}{Q}$$

g = gravitational acceleration constant = $9.81\ m/s^2$

h = head, length (m)

Q = volumetric flow rate (cm^3/s)

T_{MF} = mold filling time,

V = volume of mold cavity (m^3)

Example 1

- A mold sprue is 20 cm long, and the cross-sectional area at its base is 2.5 cm^2. The sprue feeds a horizontal runner leading into a mold cavity whose volume is 1560 cm^3. Determine:

(a) Velocity of the molten metal at the base of the sprue

(b) Volume flow rate

(c) Time to fill the mold

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Solution 1

(a) The velocity of the flowing metal at the base of the sprue is given by

$$v = \sqrt{2gh} = \sqrt{2 \times 981 \times 20} = 198.1\ cm/s$$

(b) The volumetric flow rate is given by

$$Q = vA = (2.5cm^2)(198.1cm/s) = 495\ cm^2/\text{s}$$

(c) Time required to fill a mold cavity of 1560 cm^3 at this flow rate is

$$T_{MF} = \frac{1560}{495} = 3.2\ s$$

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Example 2

A cylindrical riser must be designed for a sand-casting mold. The casting itself is a steel rectangular plate with dimensions 7.5 cm 12.5 cm 2.0 cm. Previous observations have indicated that the total solidification time (T_{TS}) for this casting = 1.6 min. The cylinder for the riser will have a diameter-to-height ratio=1.0. Determine the dimensions of the riser so that its T_{TS}=2.0 min.

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Solution 2

Solution: First determine the V/A ratio for the plate. Its volume $V = 7.5 \times 12.5 \times 2.0 = 187.5\,\text{cm}^3$, and its surface area $A = 2(7.5 \times 12.5 + 7.5 \times 2.0 + 12.5 \times 2.0) = 267.5\,\text{cm}^2$. Given that $T_{TS} = 1.6$ min, we can determine the mold constant C_m from Eq. (10.7), using a value of $n = 2$ in the equation.

$$C_m = \frac{T_{TS}}{(V/A)^2} = \frac{1.6}{(187.5/267.5)^2} = 3.26\,\text{min/cm}^2$$

Next we must design the riser so that its total solidification time is 2.0 min, using the same value of mold constant. The volume of the riser is given by

$$V = \frac{\pi D^2 h}{4}$$

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Solution 2 cont.

and the surface area is given by

$$A = \pi Dh + \frac{2\pi D^2}{4}$$

Since we are using a D/H ratio $= 1.0$, then $D = H$. Substituting D for H in the volume and area formulas, we get

$$V = \pi D^3/4$$

and

$$A = \pi D^2 + 2\pi D^2/4 = 1.5\pi D^2$$

Thus the V/A ratio $= D/6$. Using this ratio in Chvorinov's equation, we have

$$T_{TS} = 2.0 = 3.26\left(\frac{D}{6}\right)^2 = 0.09056\,D^2$$

$$D^2 = 2.0/0.09056 = 22.086\,\text{cm}^2$$

$$D = 4.7\,\text{cm}$$

Since $H = D$, then $H = 4.7\,\text{cm}$ also.

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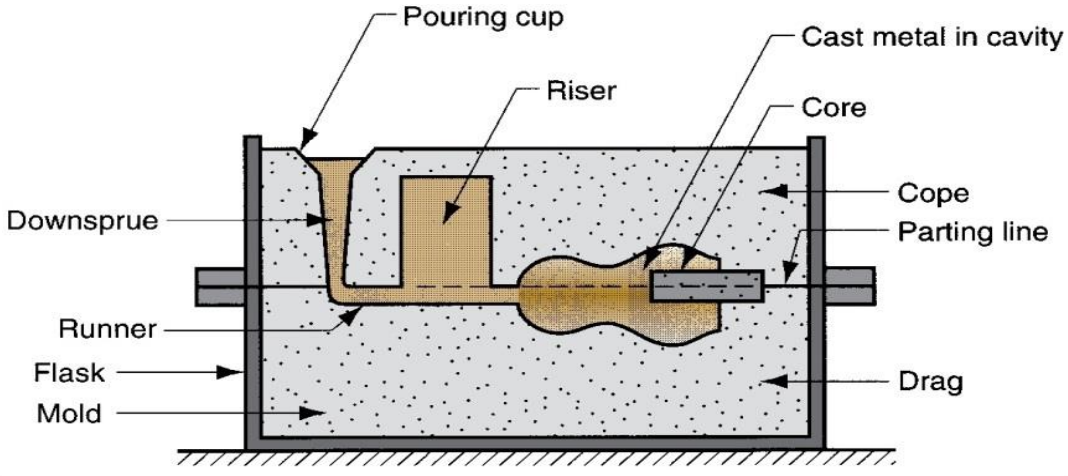
Riser Design

- Riser is waste metal that is separated from the casting and re-melted to make more castings.
- To minimize waste in the unit operation, it is desirable for the volume of metal in the riser to be a minimum.
- Since the shape of the riser is normally designed to maximize the V/A ratio, this allows riser volume to be reduced to the minimum possible value.

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Riser Design

- A riser is used in a sand-casting mold to feed liquid metal to the casting during freezing in order to compensate for solidification shrinkage.
- To function, the riser must remain molten until after the casting solidifies.
- Chvorinov's rule can be used to compute the size of a riser that will satisfy this requirement.



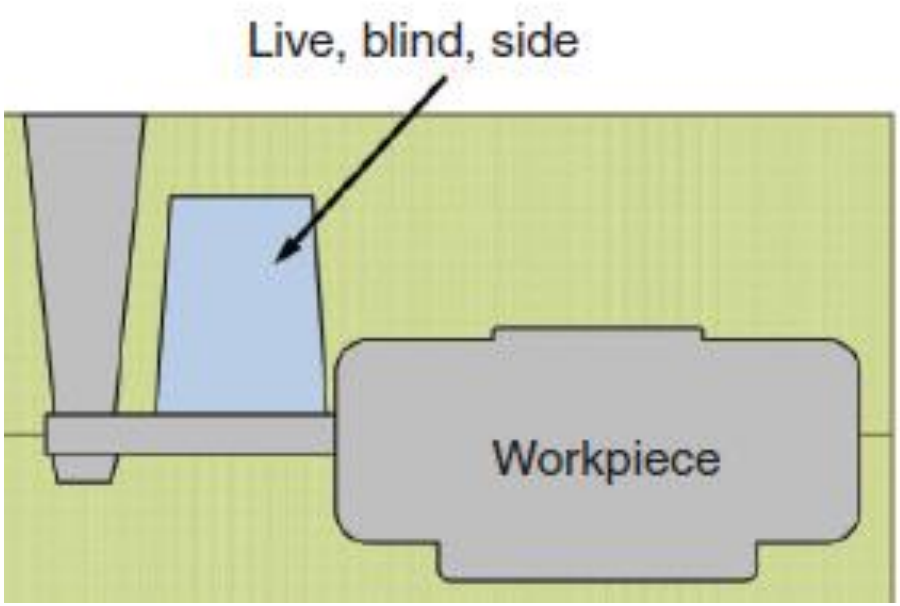
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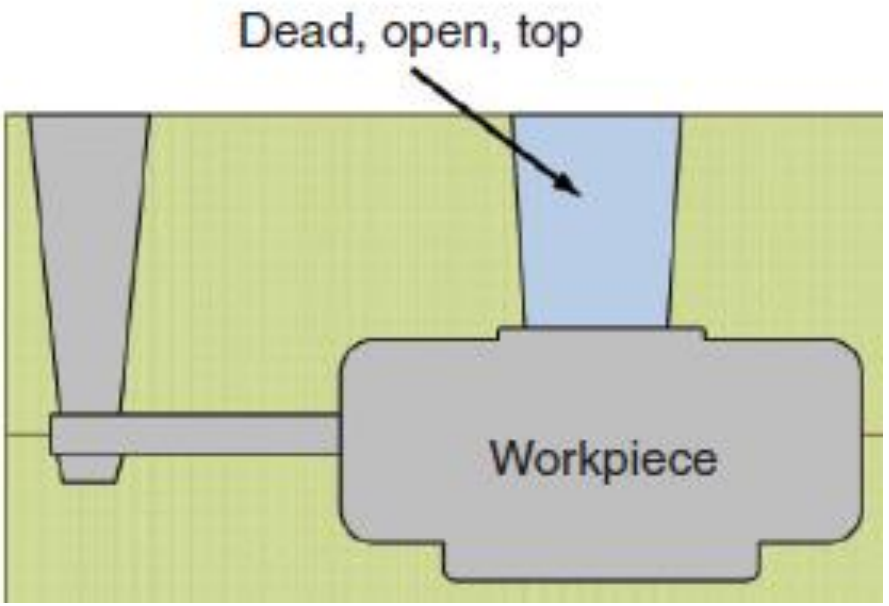
Riser Design

- Risers can be designed in different forms.
- A **side riser** is attached to the side of the casting by means of a small channel.
- A **top riser** is one that is connected to the top surface of the casting.
- Risers can be **open or blind**.
- **An open riser** is exposed to the outside at the top surface of the cope. This has the disadvantage of allowing more heat to escape, promoting faster solidification.
- A **blind riser** is entirely enclosed within the mold.

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Riser Design



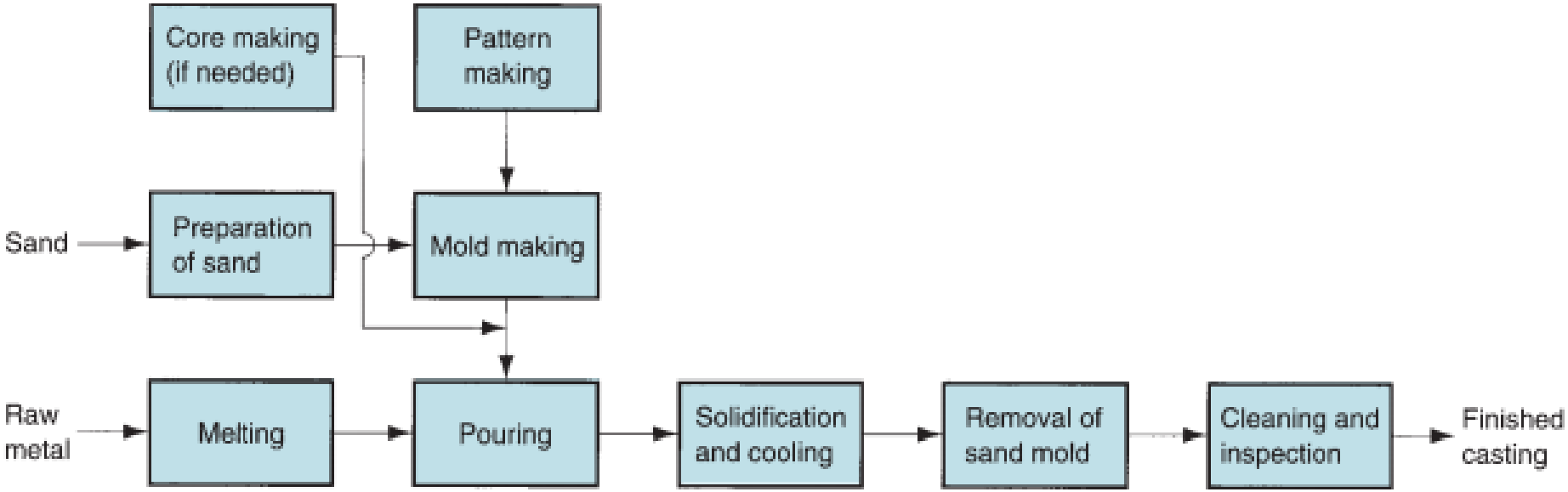


Metal Casting Processes

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Sand Casting

- Sand casting is the most widely used casting process, accounting for a significant majority of the total tonnage cast.
- Nearly all casting alloys can be sand cast.
- Sand casting is one of the few processes that can be used for metals with high melting temperatures, such as steels, nickels, and titaniums.



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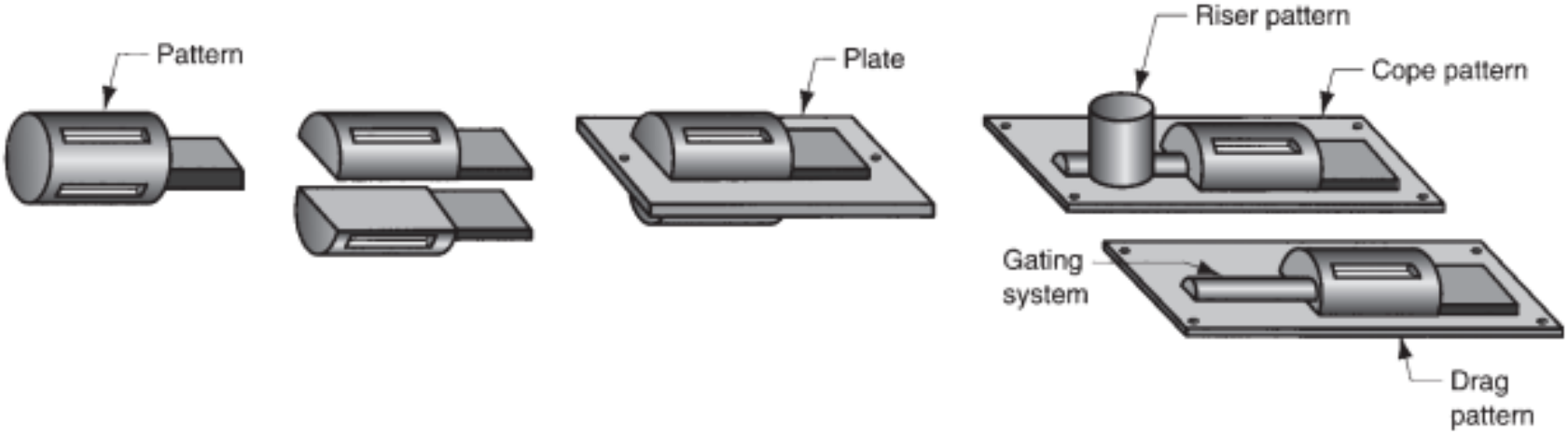
Patterns

- A **pattern** is a full-sized model of the part, enlarged to account for shrinkage and machining allowances in the final casting.
- Materials used to make patterns include **wood, plastics, and metals.**
- **Wood** is a common pattern material because it is easily shaped.
- **Metal patterns** are more expensive to make, but they last much longer.
- **Plastics** represent a compromise between wood and metal.
- Selection of the appropriate pattern material depends on the total quantity of castings to be made.

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Patterns

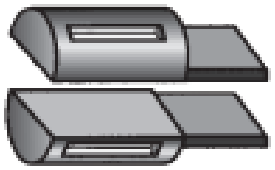
- Patterns may be classified into solid pattern, split pattern, match-plate patterns and cope-and-drag patterns.
- **Solid pattern** is the easiest to fabricate. It has same geometry as the casting, adjusted in size for shrinkage and machining.
- Solid patterns are not easy to use in making the sand mold.
- Solid patterns are generally limited to very low production quantities.



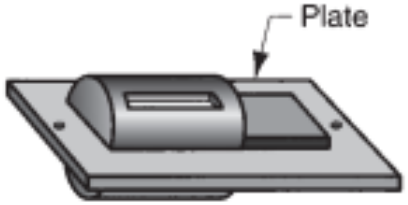
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Patterns

- **Split patterns** consist of two pieces, dividing the part along a plane coinciding with the parting line of the mold.
- Split patterns are appropriate for complex part geometries and moderate production quantities.
- In **match-plate patterns**, the two pieces of the split pattern are attached to opposite sides of a wood or metal plate. Holes in the plate allow the top and bottom (cope and drag) sections of the mold to be aligned accurately.



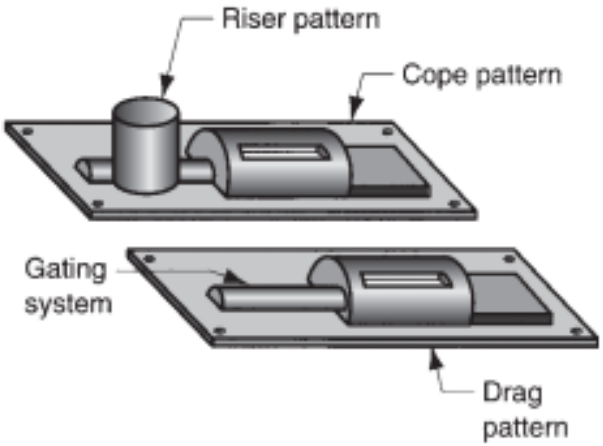
Split pattern



Match-plate pattern

Patterns

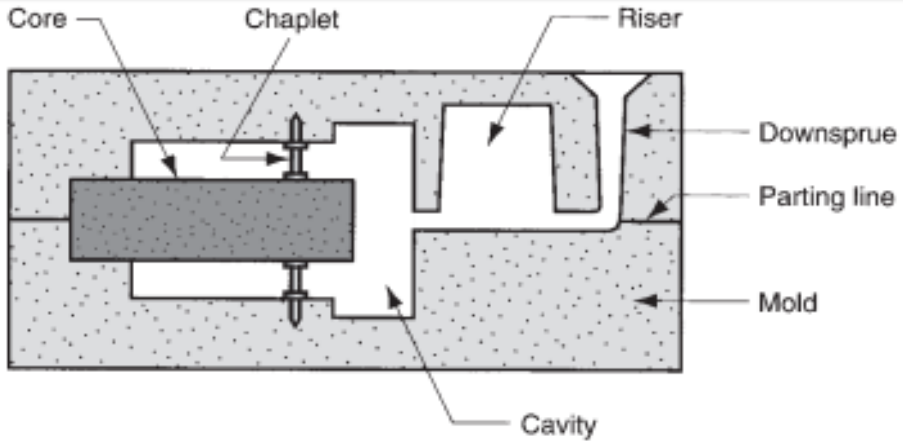
- **Cope-and-drag patterns** are similar to match-plate patterns except that split pattern halves are attached to separate plates, so that the cope and drag sections of the mold can be fabricated independently, instead of using the same tooling for both.
- For higher production quantities, match-plate patterns or cope-and-drag patterns are used.



Cope and drag pattern

Cores

- A **core** is a full-scale model of the interior surfaces of the part which is inserted into the mold cavity prior to pouring, so that the molten metal will flow and solidify between the mold cavity and the core to form the casting's external and internal surfaces.
- The core is usually made of sand, compacted into the desired shape.
- The actual size of the core must include allowances for shrinkage and machining. Supports, called chaplets, are used to hold the part in position in the mold cavity during pouring.



Molds and mold making

- Foundry sands are silica (SiO_2) or silica mixed with other minerals.
- The sand should possess good refractory properties ie. capacity to stand up under high temperatures without melting or otherwise degrading.
- Other important features of the sand include grain size, distribution of grain size in the mixture, and shape of the individual grains.
- Small grain size provides a better surface finish on the cast part, but large grain size is more permeable (to allow escape of gases during pouring).

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Molds and mold making

- In making the mold, the grains of sand are held together by a mixture of **water and bonding clay.**
- A typical mixture (by volume) is **90% sand, 3% water, and 7% clay.**
- Other bonding agents can be used in place of clay, including organic resins (e.g., phenolic resins) and inorganic binders (e.g., sodium silicate and phosphate).
- Besides sand and binder, additives are sometimes combined with the mixture to enhance properties such as strength and/or permeability of the mold.

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Indicators of a quality sand mold

- **Strength**: Ability of the mold to maintain its shape and resist erosion caused by the flow of molten metal. It depends on grain shape, adhesive qualities of the binder, and other factors.
- **Permeability**: Ability of the mold to allow hot air and gases from the casting operation to pass through the voids in the sand.
- **Thermal stability**: Ability of the sand at the surface of the mold cavity to resist cracking and buckling upon contact with the molten metal.
- **Collapsibility:** Ability of the mold to give way and allow the casting to shrink without cracking the casting; it also refers to the ability to remove the sand from the casting during cleaning; and
- **Reusability**: Ability of the sand from the broken mold to be reused to make other molds.

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Bouyancy in sand casting

- The casting operation involves pouring, solidification, and cooling of the cast part.
- One of the hazards during pouring is that the buoyancy of the molten metal will displace the core.
- Buoyancy results from the weight of molten metal being displaced by the core, according to Archimedes' principle.
- The force tending to lift the core is equal to the weight of the displaced liquid less than the weight of the core itself.
- This can be expressed mathematically as

$$F_b = W_m - W_c$$

Where F_b is the buoyancy force; W_m is the weight of molten metal displaced and W_c is weight of the core.

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Example 3

A sand core has a volume of $1875\ \text{cm}^3$ and is located inside a sand mold cavity. Determine the buoyancy force tending to lift the core during pouring of molten lead into the mold. Take density of sand core and lead to be 1.6 g/cm^3 and $11.3\ g/\text{cm}^3$ respectively.

Solution

Density of the sand core is 1.6 g/cm^3 .

Weight of the core is $1875 \times 1.6 = 3000\ g = 3.0\ kg.$

Density of lead is 11.3 g/cm^3.

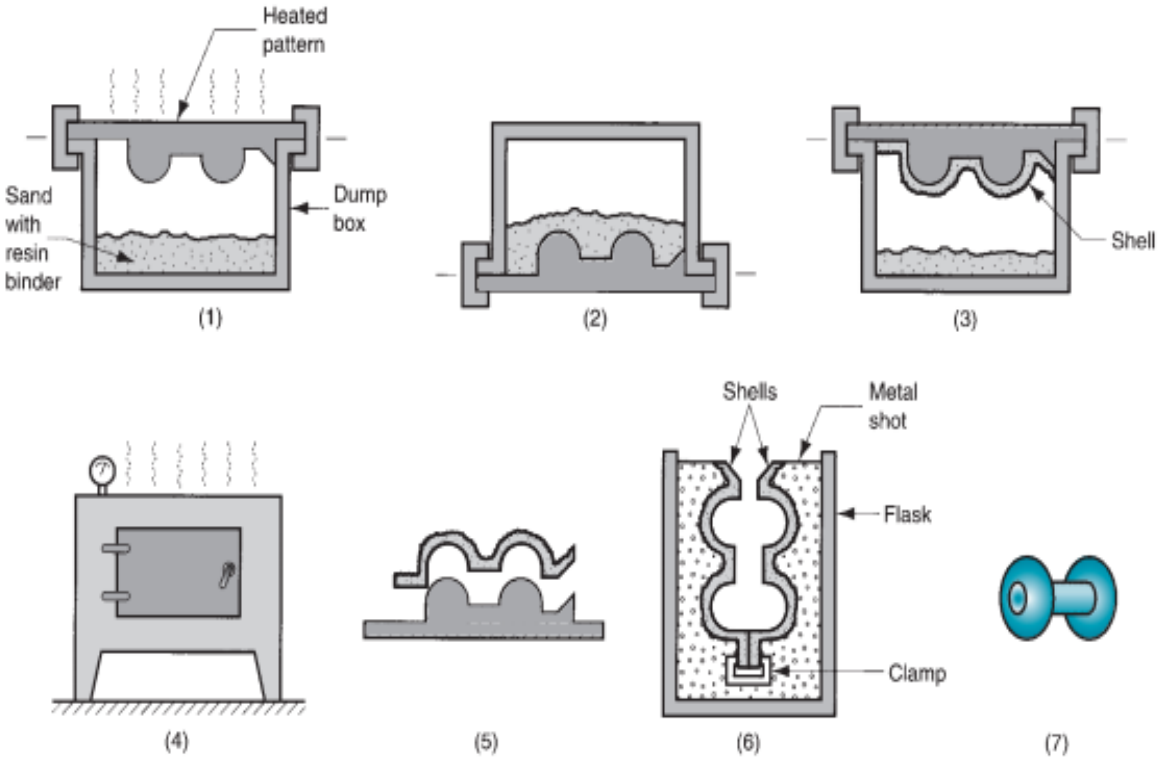
The weight of lead displaced by the core is
$1875 \times 11.3 = 21{,}188\ g = 21.19 kg.$

The difference $= 21.19 - 3.0 = 18.19\ kg.$

Given that 1 kg = 9.81 N, the buoyancy force is therefore
$F_b = 9.81 \times 18.19 = 178.4\ N$

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Shell molding



Steps in shell molding

(1) Match-plate or cope-and-drag metal pattern is heated and placed over a box containing sand mixed with thermosetting resin.

(2) Box is inverted so that sand and resin fall onto the hot pattern, causing a layer of the mixture to partially cure on the surface to form a hard shell

(3) Box is repositioned so that loose, uncured particles drop away

(4) sand shell is heated in oven for several minutes to complete curing

(5) shell mold is stripped from the pattern removed.

(6) two halves of the shell mold are assembled, supported by sand or metal shot in a box, and pouring is accomplished.

(7) The finished casting with sprue.

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Advantages of shell molding

- The surface of the shell mold cavity is smoother than a conventional green-sand mold, and this smoothness permits easier flow of molten metal during pouring and better surface finish on the final casting.
- Finishes of 2.5 μm can be obtained.
- Good dimensional accuracy is also achieved, with tolerances of $\pm$0.25 mm possible on small-to-medium-sized parts.
- The good finish and accuracy often precludes the need for further machining.
- Collapsibility of the mold is generally sufficient to avoid tearing and cracking of the casting.

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Disadvantages of shell molding

- More expensive metal pattern than the corresponding pattern for green-sand molding.
- The expensive metal pattern makes it an impractical option for small quantities of parts.

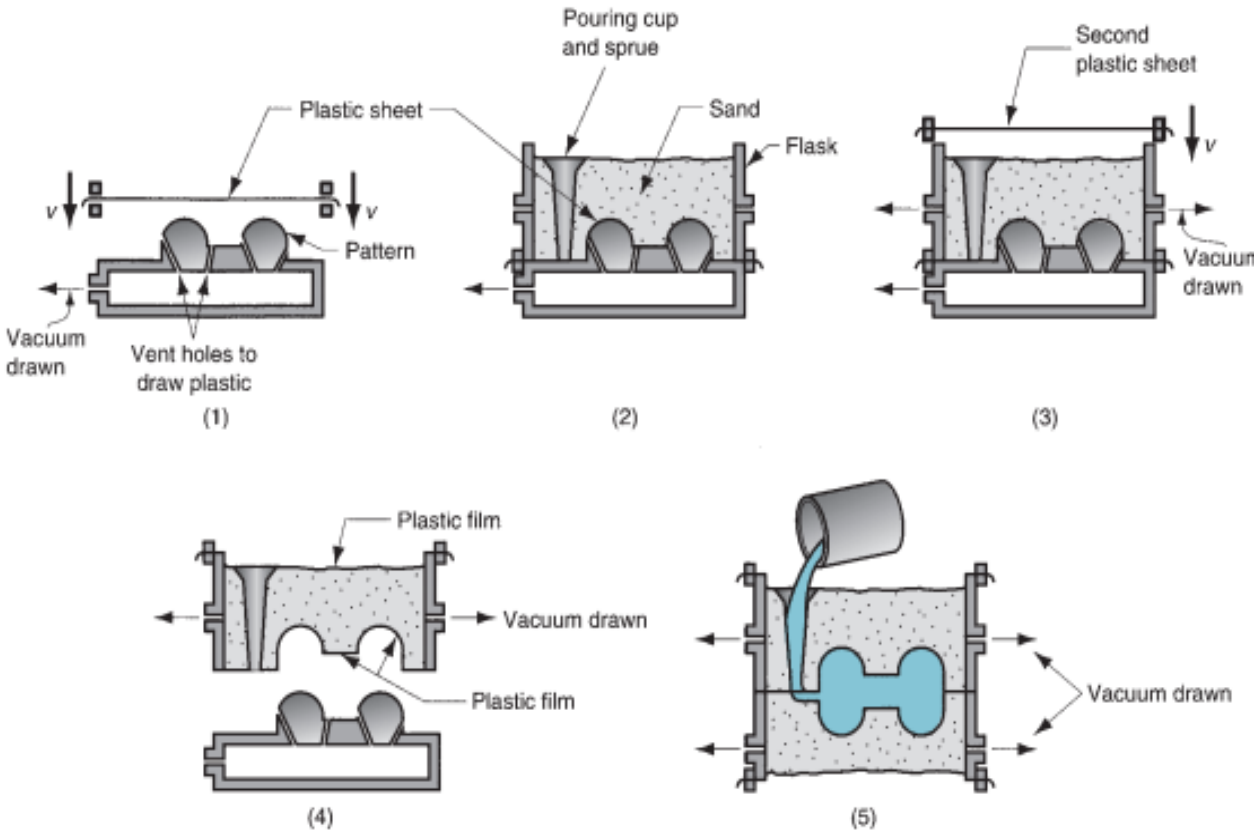
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Vacuum molding (V-process)

- Uses a sand mold held together by vacuum pressure rather than by a chemical binder.
- The term vacuum in this process refers to the making of the mold rather than the casting operation itself.
- Because no binders are used, the sand is readily recovered in vacuum molding.
- Also, the sand does not require extensive mechanical reconditioning normally done when binders are used in the molding sand.
- Since no water is mixed with the sand, moisture related defects are absent from the product.
- Disadvantages of the V-process are that it is relatively slow and not readily adaptable to mechanization.

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Steps in vacuum molding (V-process)



(1) À thin sheet of pre-heated plastic is drawn over a match-plate or cope-and-drag pattern by vacuum

(2) A specially designed flask is placed over the pattern plate and filled with sand, and a sprue and pouring cup are formed in the sand;

(3) Another thin plastic sheet is placed over the flask, and a vacuum is drawn to form a rigid mold;

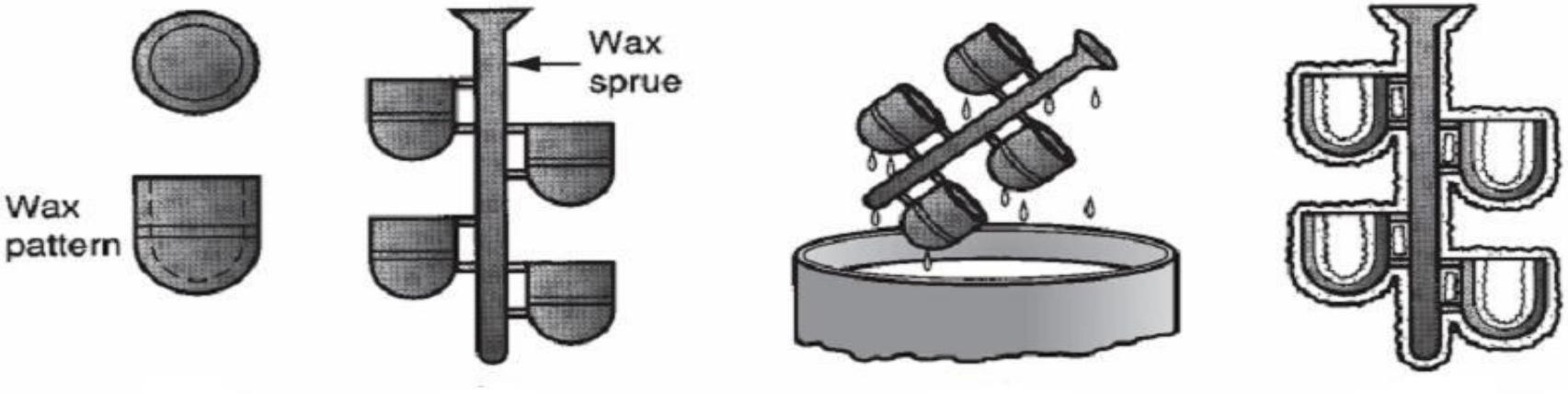
(4) The vacuum on the mold pattern is released to permit the pattern to be stripped from the mold;

(5) This mold is assembled with its matching half to form the cope and drag, and with vacuum maintained on both halves, pouring is accomplished. The plastic sheet quickly burns away on contacting the molten metal. After solidification, nearly all of the sand can be recovered for reuse

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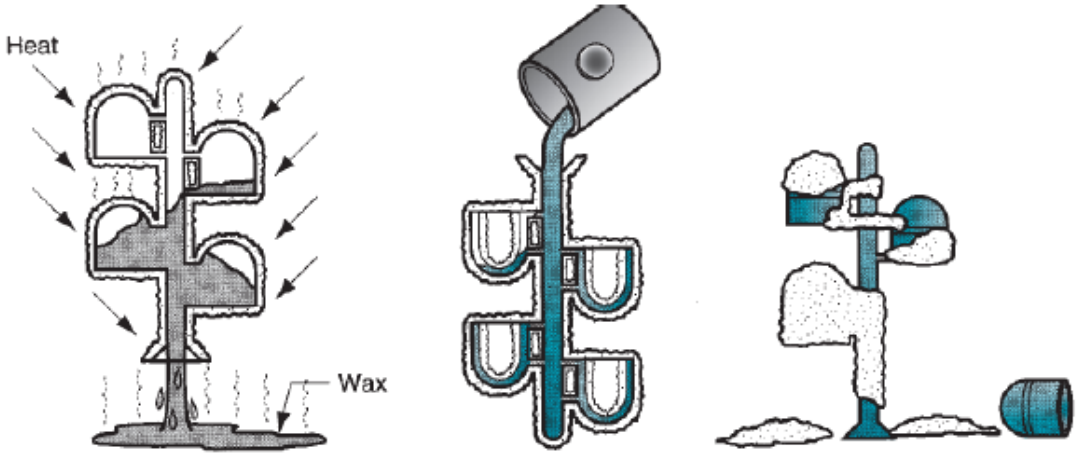
Casting Processes – Investment casting

- In this casting process, a pattern made of wax is coated with a refractory material to make the mold surface, after which the wax is melted away while pouring the molten metal. "Investment" means "to cover completely" which refers to the coating of the refractory material around the wax pattern.
- This is a precision casting process. Using this we can make castings of high accuracy with intricate details.



Investment casting

- Wax patterns are first made
- several patterns can be attached to a sprue to form a pattern tree, if required
- the pattern tree is coated with a thin layer of refractory material and later covered with thick coating to make the rigid full mold



- Heating of mold in inverted position to melt the wax and permit it to drip out of the cavity
- the mold is preheated to a high temperature so that contaminants are eliminated from the mold
- the molten metal is poured and it solidifies
- the mold is removed from the finished casting

Refractory coating

- Slurry of very fine grained silica or other refractory, in powder form, mixed with plaster to bond the mold into shape. The small grain size of the refractory material delivers smooth surface and captures the intricate depths of the wax pattern.
- Mold is allowed to dry in air for about 8 hours to harden the binder.

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Investment casting

Advantages

- Complex and intricate parts can be cast
- Tolerances of 0.075 mm are possible
- Good surface finish is possible
- In general, additional machining is not required – near net shaped part

Applications

Steels, stainless steels, high temperature alloys can be cast - Examples of parts: machine parts, blades, components for turbine engines, jewelry, dental fixtures

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Plaster Mold Casting

- Similar to sand casting, except mold is made of POP and not sand.
- To minimize contraction, curing time, reduce cracking, additives like talc and silica flour are mixed with the plaster.
- Curing time: 20 min, baking time: several hours.
- Permeability is low. This problem is solved by using a special mold composition and treatment known as the Antioch process. In this operation, about 50% of sand is mixed with the plaster, heating the mold in an autoclave, and then drying is done. Good permeability is attained by this treatment.
- Used only for Al, Mg, Cu based alloys.

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Ceramic mold casting

- Mold is made of refractory ceramic materials which can withstand high temperature than plaster.
- Ceramic molding can be used to cast steels, CI, and other high temperature alloys.

- **Disadvantage of expendable molding processes** is that for every casting a new mold is required.

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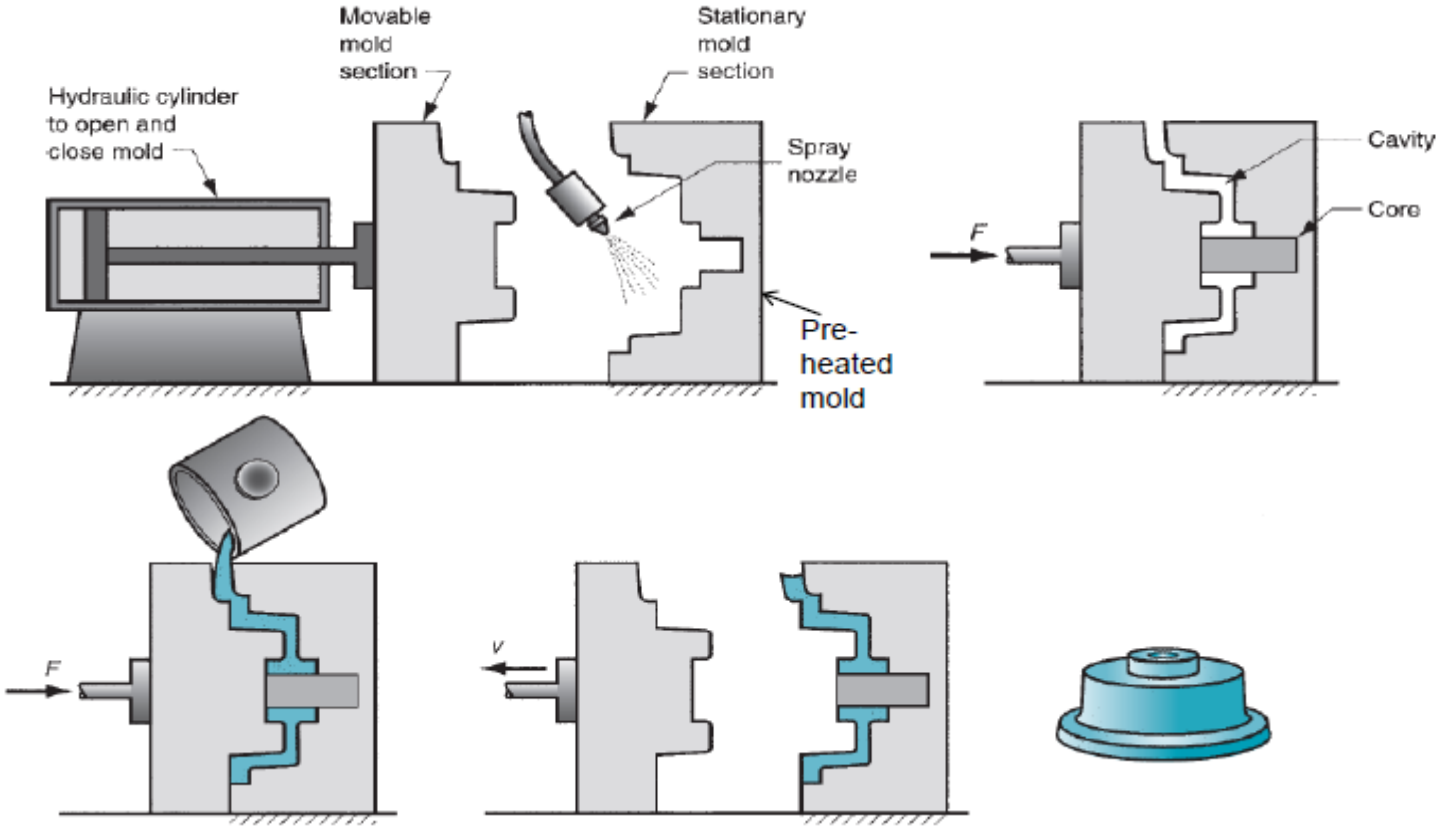
Permanent Mold Process

- Using only metal mold for casting
- Molds are generally made of steel, CI
- Materials that can be cast: Al, Mg, Cu based alloys, CI (affect the mold life, hence not used)
- Cores are also made of metal, but if sand is used then called semi permanent-mold casting

- **Advantages**: good surface finish, dimension tolerance, rapid solidification causes fine grains to form giving stronger products
- **Limitations:** restricted to simple part geometries, low melting point metals, mold cost is high. Best suitable for small, large number of parts

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Steps in Permanent Mold Casting

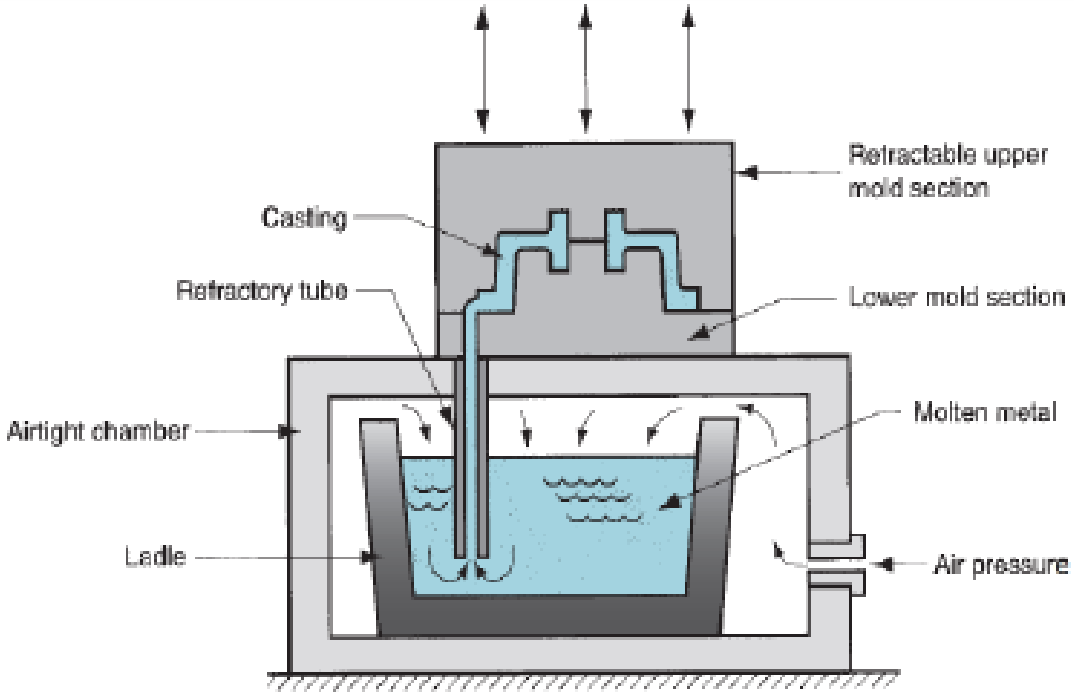


- Preheating facilitates metal flow through the gating system and into the cavity.
- The coatings aid heat dissipation and lubricate the mold surfaces for easier separation of the cast product

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Variations of Permanent mold casting (Low pressure casting)

In the earlier casting process, metal flow in mold cavity is by gravity pull, but in low pressure casting, liquid metal is forced into the cavity under low pressure, approximately 0.1 MPa, from beneath the surface so that metal flow is upward.



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Vacuum Permanent Mold Casting

- Variation of low pressure casting, but in this vacuum is used to draw the molten metal into the mold cavity.

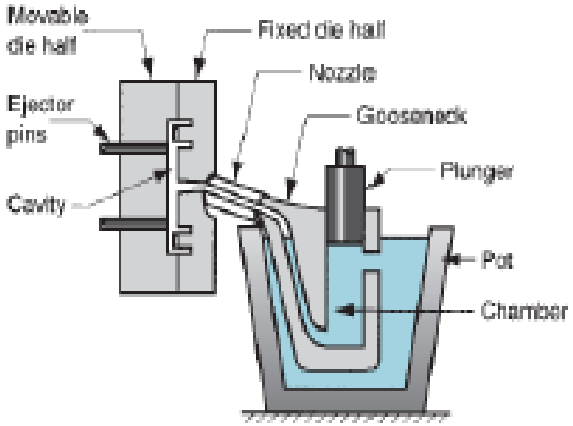
Advantage

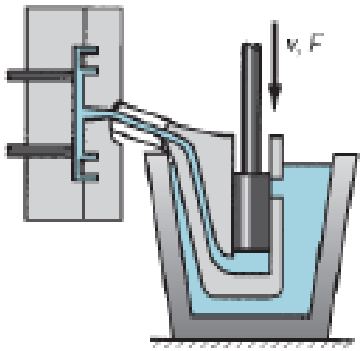
Molten metal is not exposed to air; gas porosity and oxidation defects are minimized

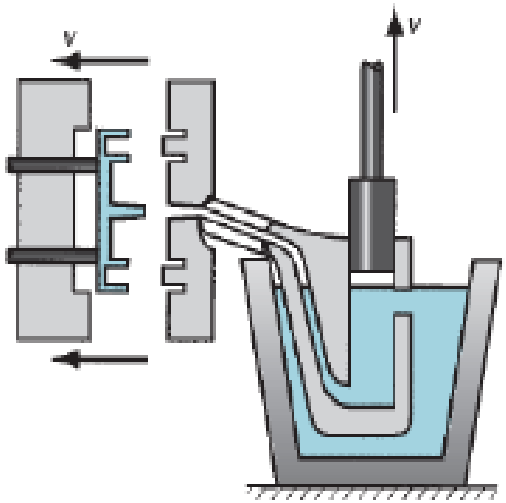
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Die Casting

- In this process, high pressure of approximately 7 to 350 MPa is used to pressurize the molten metal into die cavity. The pressure is maintained during solidification.
- **Types**: Hot chamber and cold chamber die casting









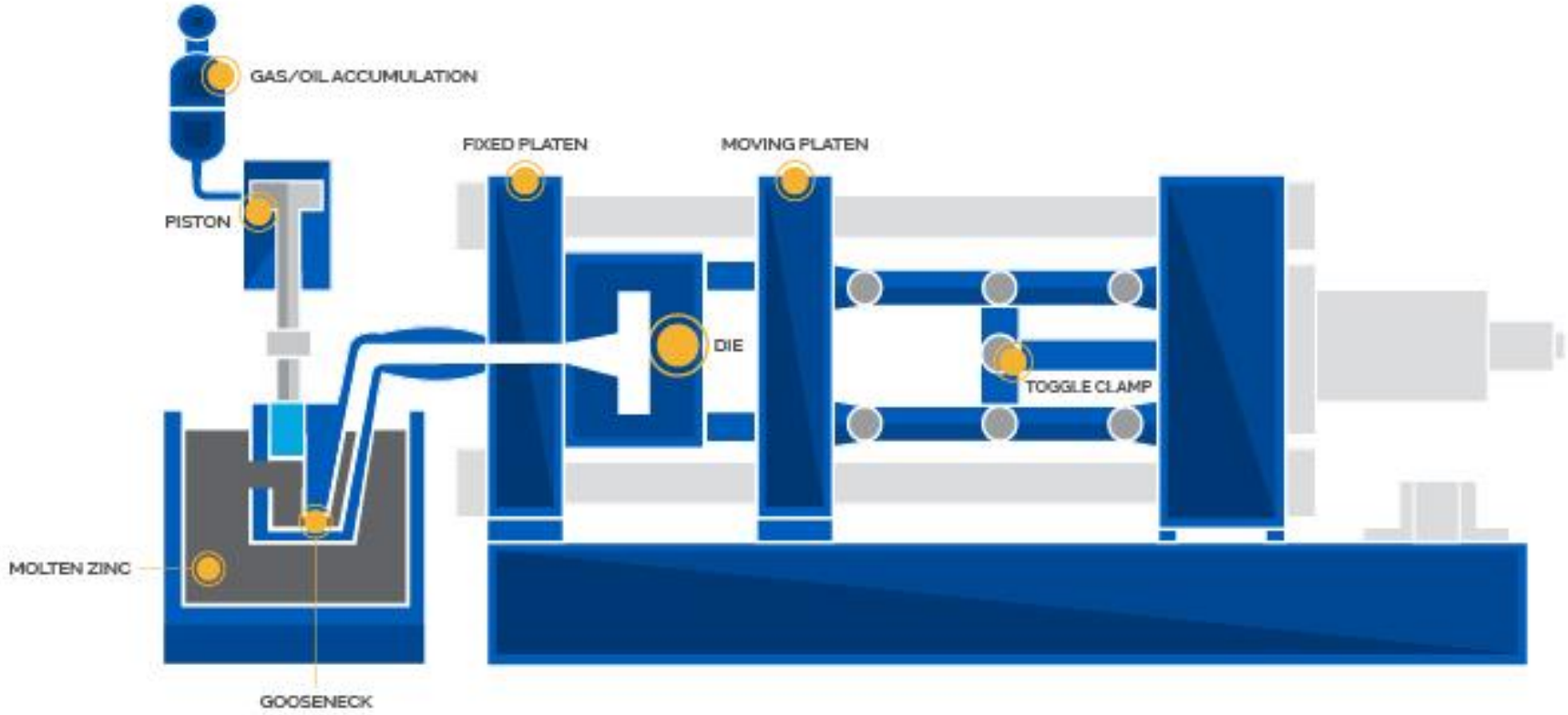
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Hot Chamber Die Casting

- Molten metal is melted in a container attached to the machine, and a piston is used to pressurize metal under high pressure into the die.
- Typical injection pressures are between 7 and 35 MPa.
- Production rate of 500 parts/hour are common.
- Injection system is submerged into the molten metal and hence pose problem of chemical attack on the machine components. Suitable for zinc, tin, lead, Mg.

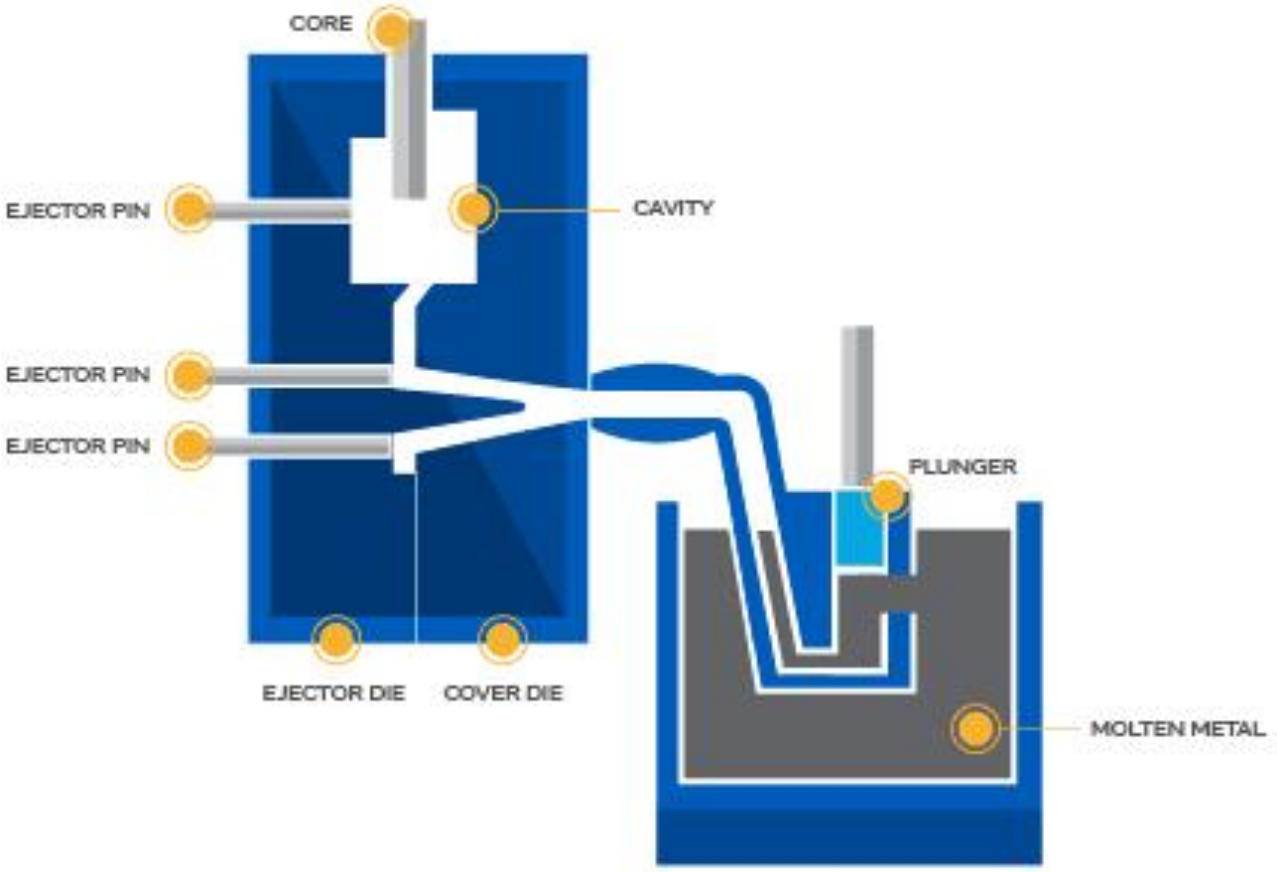
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Components of the hot chamber die casting machine



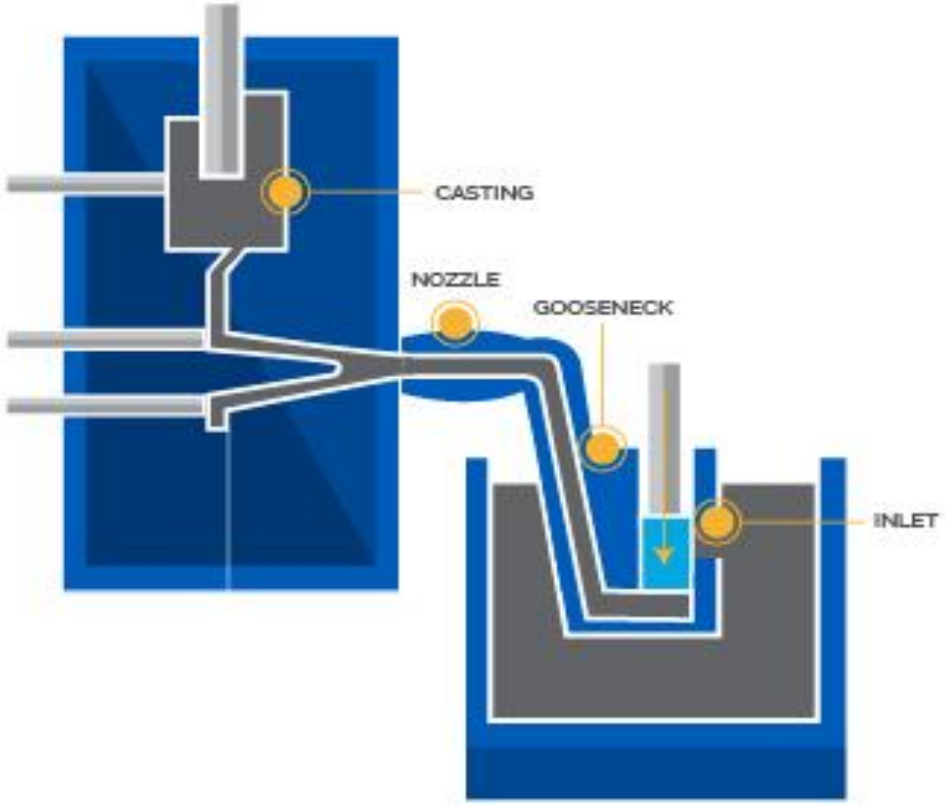
The injection mechanism of a hot chamber machine is immersed in the molten metal. The furnace is attached to the machine by a metal feed system called a gooseneck.

Components of the hot chamber die casting machine



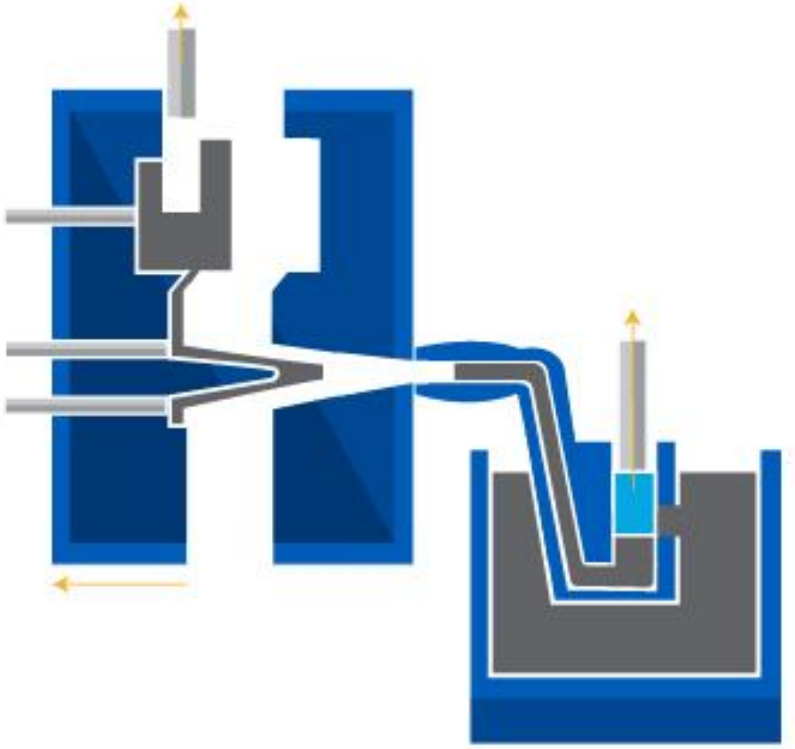
- The die is closed and the piston rises, opening the port, allowing molten metal to fill the cylinder.

Components of the hot chamber die casting machine



- Next, the plunger seals the port, pushing the molten metal through the gooseneck and nozzle into the die cavity where it is held under pressure until it solidifies.

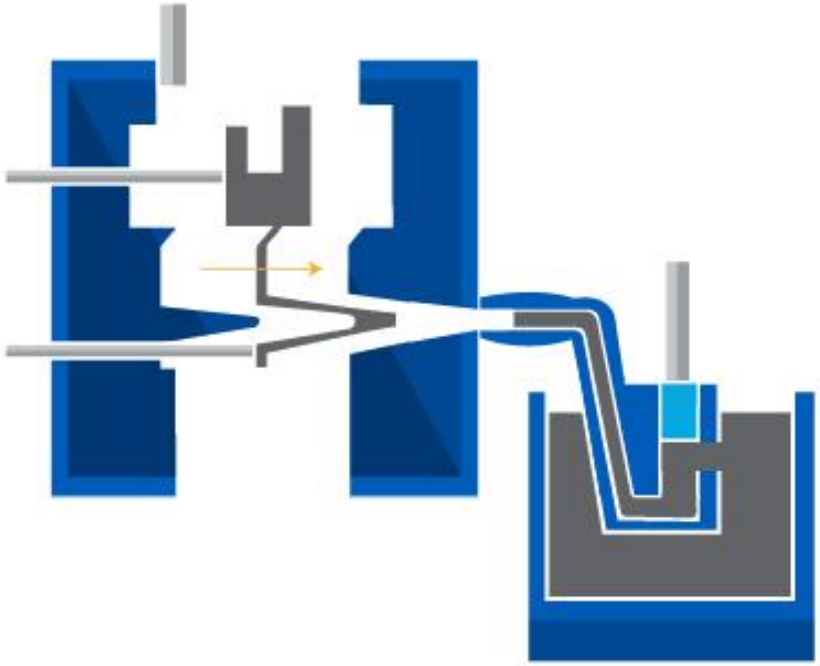
Components of the hot chamber die casting machine



- The die opens and the cores, if any, retract. The casting remains in only one die half – the ejector side. The plunger then returns, allowing residual molten metal to flow back through the nozzle and gooseneck.

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Components of the hot chamber die casting machine

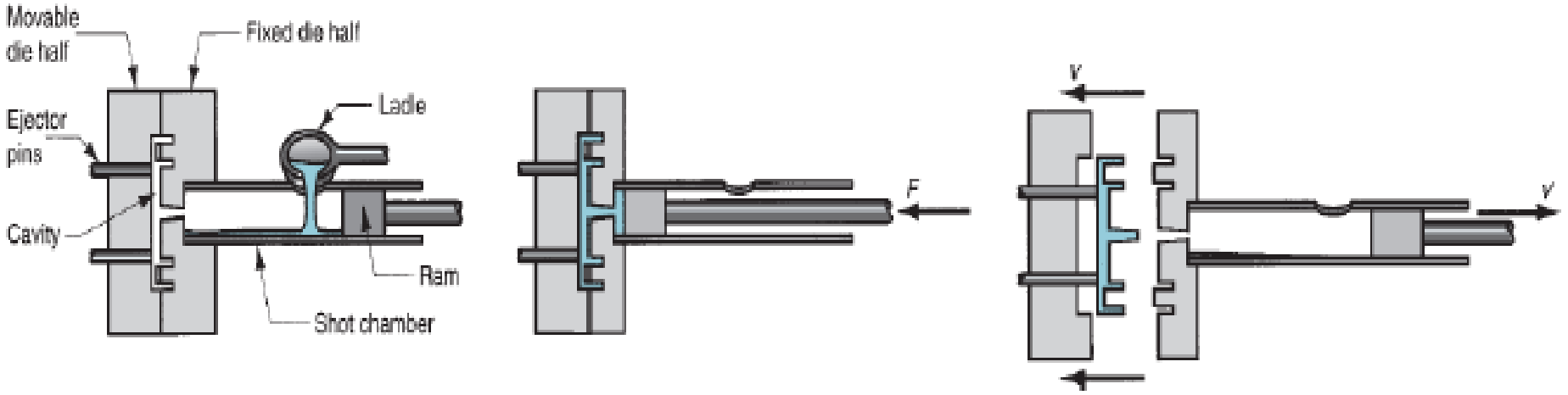


- Ejector pins push the casting out of the ejector die. As the plunger uncovers the filling hole, molten metal flows through the inlet to refill the gooseneck.

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Cold Chamber Machines

- Molten metal is poured from an external unheated container into the mold cavity and piston is used to inject the molten metal into the die cavity.
- Injection pressure: 14 to 140 MPa.
- Though it is a high production operation, it is not as fast as hot chamber machines.



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Advantages of Die Casting

- Die casting molds are made of tool steel, mold steel, maraging steels. Tungsten and molybdenum with good refractory qualities are also used for die cast steel, CI.
- High production rates and economical
- Close tolerances possible of the order of ±0.076 mm
- Thin section with 0.5 mm can be made
- Small grain size and good strength casting can be made because of rapid cooling

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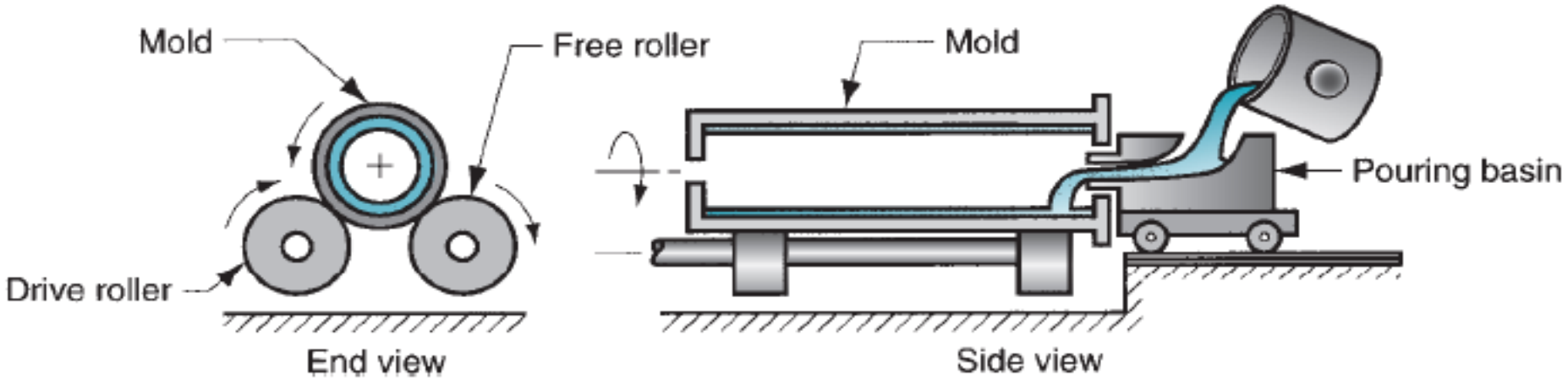
Centrifugal Casting

- In this method, the mold is rotated at high speed so that the molten metal is distributed by the centrifugal force to the outer regions of the die cavity
- **Types** : true centrifugal casting, semi centrifugal casting

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True Centrifugal Casting

- Molten metal is poured into a rotating mold to produce a tubular part (pipes, tubes, bushings, and rings).
- Molten metal is poured into a horizontal rotating mold at one end. The high-speed rotation results in centrifugal forces that cause the metal to take the shape of the mold cavity.
- The outside shape of the casting can be non-round, but inside shape of the casting is perfectly round, due to the radial symmetry with respect to forces.



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Horizontal Centrifugal Casting

- Orientation of the mold can be **horizontal or vertical**
- Centrifugal force $(F) = \frac{mv^2}{R}$

Where F is force (N), m is mass (kg), v is velocity (m/s), R is inner radius of mold (m)

- G-factor (GF) is the ratio of centrifugal force to weight

$$GF = \frac{\frac{mv^2}{R}}{mg} = \frac{v^2}{Rg}$$

- However, $v = \frac{2\pi RN}{60}$; $N = \frac{30}{\pi}\sqrt{\frac{2g(GF)}{D}}$

Where N is rotational speed (rev/min) and

D is inner diameter of mold (m)

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Horizontal Centrifugal Casting

- For horizontal centrifugal casting, GF is equal to 60 to 80
- If the G-factor is very less, because of the reduced centrifugal force, the liquid metal will not remain forced against the mold wall during the upper half of the circular path but will go into the cavity. This means that slipping occurs between the molten metal and the mold wall, which indicates that rotational speed of the metal is less than that of the mold.

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Vertical Centrifugal Casting

- In this because of the effect of gravity acting on the liquid metal, casting wall will be thicker at the base than at the top. The difference in inner and outer radius can be related to speed of rotation as,

$$N = \frac{30}{\pi}\sqrt{\frac{2gL}{R_{it}^2 - R_{ib}^2}}$$

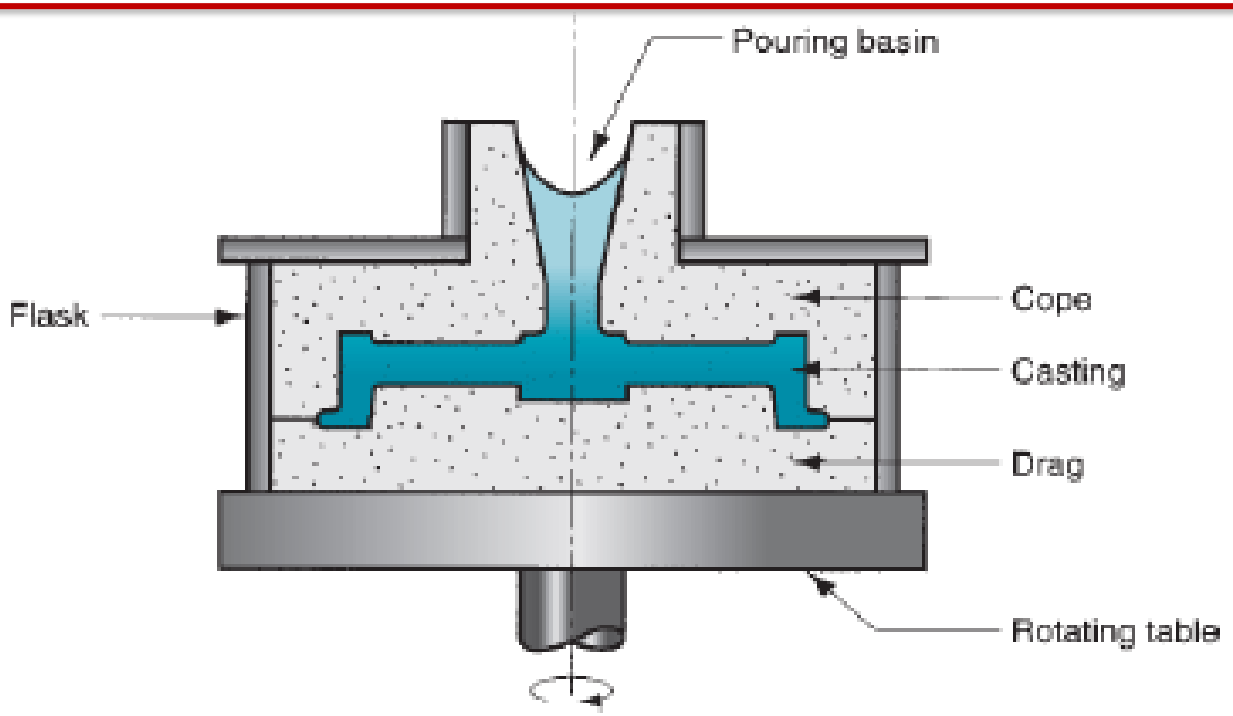
Where L - vertical length of the casting (m)

Rit - inner radius at the top of the casting (m)

Rib - inner radius at the bottom of the casting (m)

Semi-centrifugal Casting

- In this process, centrifugal force is used to produce non-tubular parts (solid), and not tubular parts. GF will be around 15 by controlling the rotation speed. Molds are provided with riser at the center.

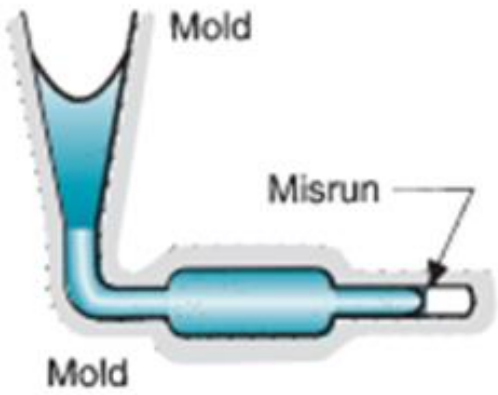


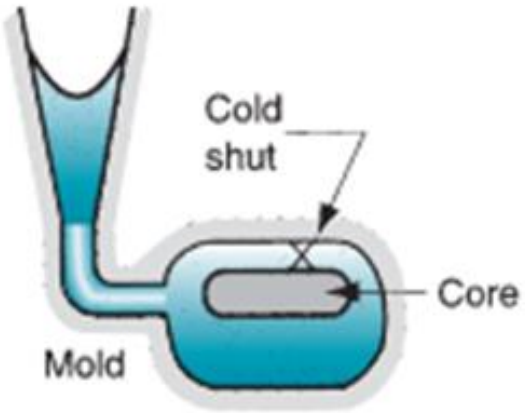
Defects in Casting

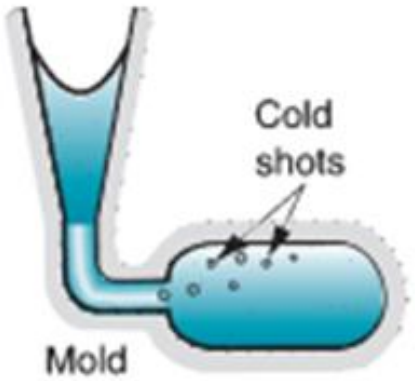
- Defects can mostly be minimized or eliminated by
  - ✓Proper design
  - ✓Preparation of molds
  - ✓Control of pouring procedures

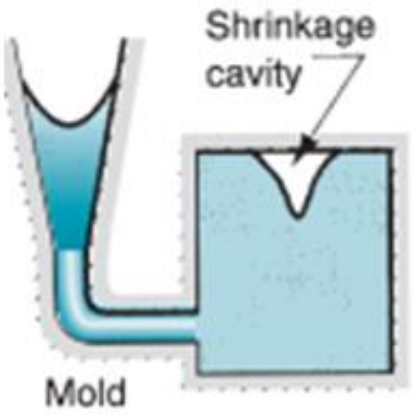
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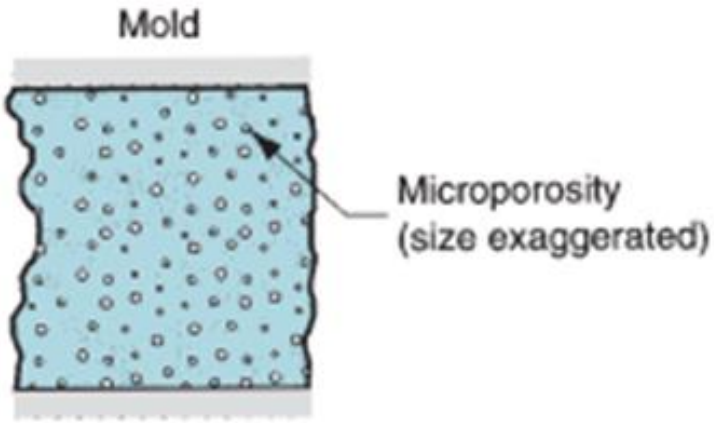
Defects in Casting

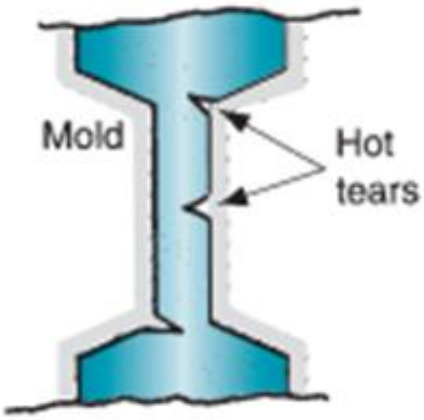




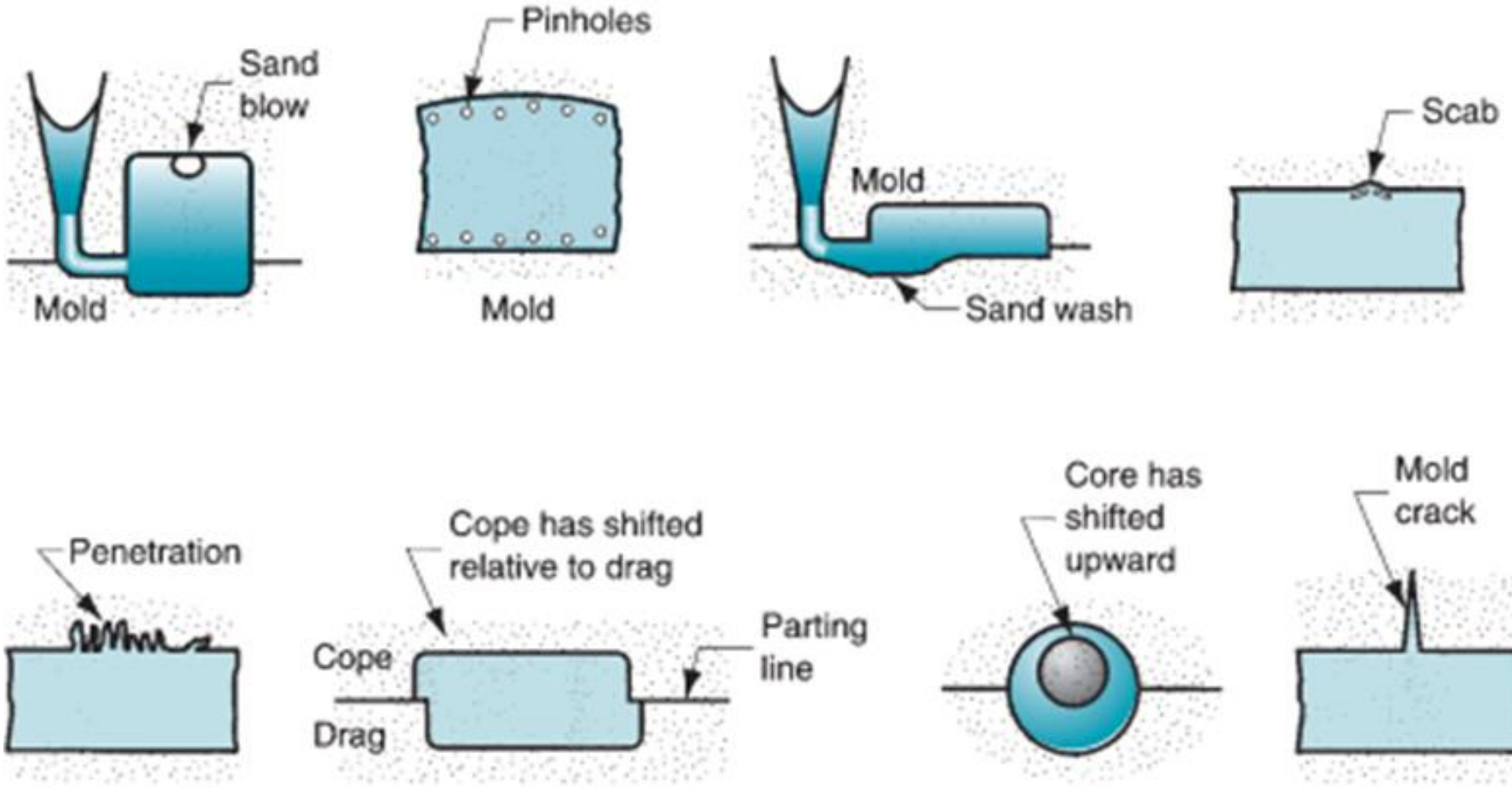








Defects in sand casting

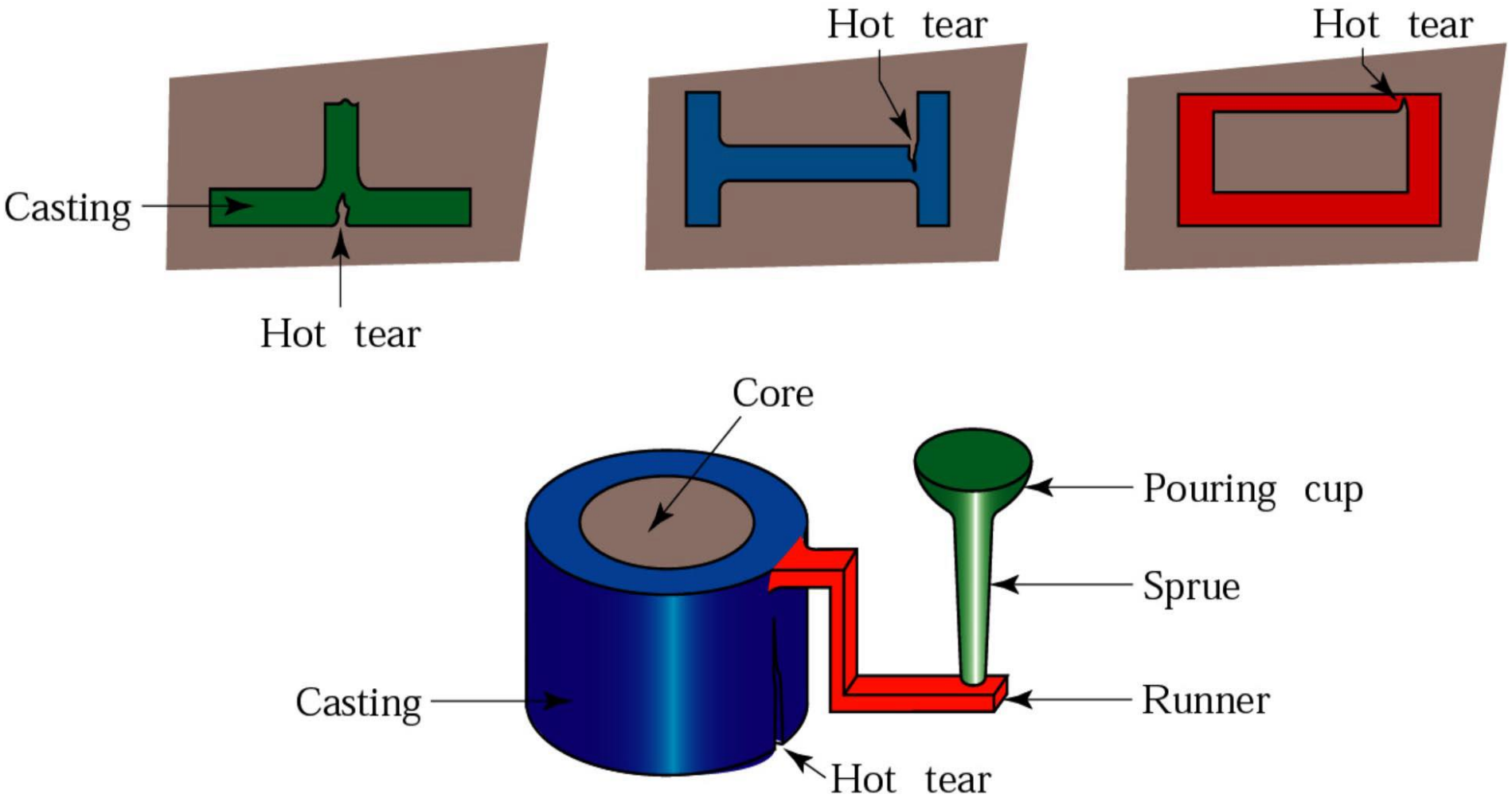


Defects in Casting – Hot tears

- These defects occur because the casting cannot shrink freely during cooling, owing to constraints in various portions of the molds and cores.
- Exothermic (heat-producing) compounds may be used (as exothermic padding) to control cooling at critical sections to avoid hot tearing.

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Defects in Casting – Hot tears



Casting of Plastics

- Some thermoplastics (such as nylons and acrylics) and thermosetting plastics (epoxies, phenolics, polyurethanes, and polyester) can be cast into a variety of shapes using either rigid or flexible molds.
- Compared with other methods of processing plastics, casting is a slow, but simple and inexpensive, process.
- However, the polymer must have sufficiently low viscosity in order to flow easily into the mold.
- Typical parts cast are gears (especially nylon), bearings, wheels, thick sheets, lenses, and components requiring resistance to abrasive wear.

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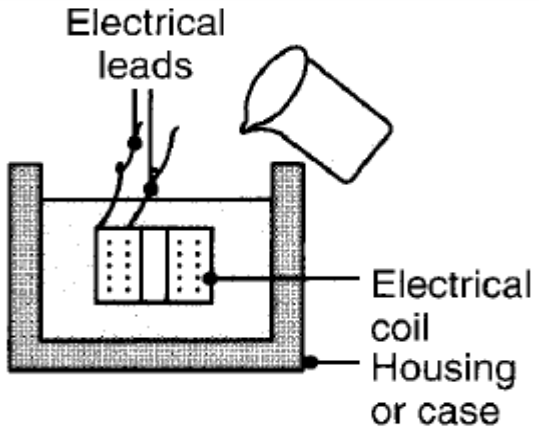
Casting of Plastics

- In the basic conventional casting of thermoplastics, a mixture of monomer, catalyst, and various additives (activators) is heated to above its melting point, Tm, and poured into the mold.
- The part is formed after polymerization takes place at ambient pressure. Degassing may be necessary for product integrity.
- Intricate shapes can be produced using flexible moldss, Which are then peeled off (in a manner similar to using rubber gloves) and reused.
- As with metals, thermoplastics may be cast continuously, with the polymer carried over continuous stainless-steel belts and polymerized by external heat.

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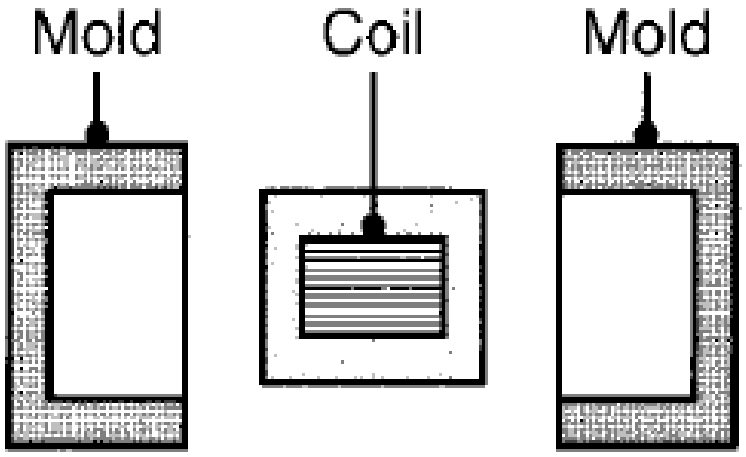
Potting and Encapsulation

- As a variation of casting that is important, particularly to the electrical and electronics industry, potting and encapsulation involve casting the plastic material (typically a liquid resin, such as expoxy) around an electrical component (such as a transformer) to embed it in the plastic.
- Potting is carried out in a housing or case, which becomes an integral part of the component and fixes it in position.



Potting and Encapsulation

- In encapsulation, the component is coated with a layer of the plastic, surrounding it completely and then solidifying.



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Potting and Encapsulation

- In both of these processes, the plastic material can serve as a dielectric (non-conductor); consequently, it must be free of moisture and porosity, which would require processing in a vacuum.
- Mold materials may be metal, glass, or various polymers.
- Small structural members (such as hooks, studs, and similar parts) may be encapsulated partially by dipping them in a hot thermoplastic using polymers of various colours.

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Product design considerations

- **Geometric simplicity**. Avoiding unnecessary complexities simplifies mold making, reduces the need for cores, and improves the strength of the casting.
- **Corners.** Avoid sharp corners and angles. Design generous fillets on inside corners, and blend sharp edges.
- **Section thicknesses**. Section thicknesses should be uniform in order to avoid shrinkage cavities.
- **Draft.** Part sections that project into the mold should have a draft or taper. Drafts facilitate removal of the pattern from the mold.
- **Use of cores**. Minor changes in part design can reduce the need for coring.

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Product design considerations

- **Dimensional tolerances**. Significant differences in dimensional accuracies can be achieved in castings.
- **Surface finish**. Typical surface roughness achieved in sand casting is around 6 mm (250 μ-in). Similarly poor finishes are obtained in shell molding, while plaster-mold and investment casting produce much better roughness values.
- **Machining allowances.** Additional material, called the machining allowance, is left on casting for machining surfaces where necessary to achieve sufficient tolerances and required dimensions. Almost all sand castings must be machined to some extent in order for the part to be made functional. Typical machining allowances for sand castings range between 1.5 mm and 3 mm.

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Summary of Casting Processes

Process Advantages	Advantages	Limitations
Sand	Almost any metal cast; no limit to size, shape or weight; low tooling cost.	Some finishing required; somewhat coarse finish; wide tolerances .
Shell mold	Good dimensional accuracy and surface finish; high production rate.	Part size limited; expensive patterns and equipment required.
Expendable pattern	Most metals cast with no limitto size; complex shapes	Patterns have low strength and can be costly for low quantities
Plaster mold	Intricate shapes; good dimensional accuracy and finish; low porosity.	Limited to nonferrous metals; limited size and volume of production; mold making time relatively long.
Ceramic mold	Intricate shapes ; close tolerance parts; good surface finish.	Limited size.

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Summary of Casting Processes

Process Advantages	Advantages	Limitations
Investment casting	Intricate shapes ; excellent surface finish and accuracy; almost any metal cast.	Part size limited; expensive patterns , molds , and labour.
Permanent mold	Good surface finish and dimensional accuracy; low porosity; high production rate .	High mold cost; limited shape and intricacy; not suitable for high melting point metals .
Die Casting	Excellent dimensional accuracy and surface finish; high production rate .	Die cost is high; part size limited; usually limited to nonferrous metals ; long lead time.
Centrifugal casting	Large cylindrical parts with good quality; high production rate.	Equipment is expensive; part shape limited.

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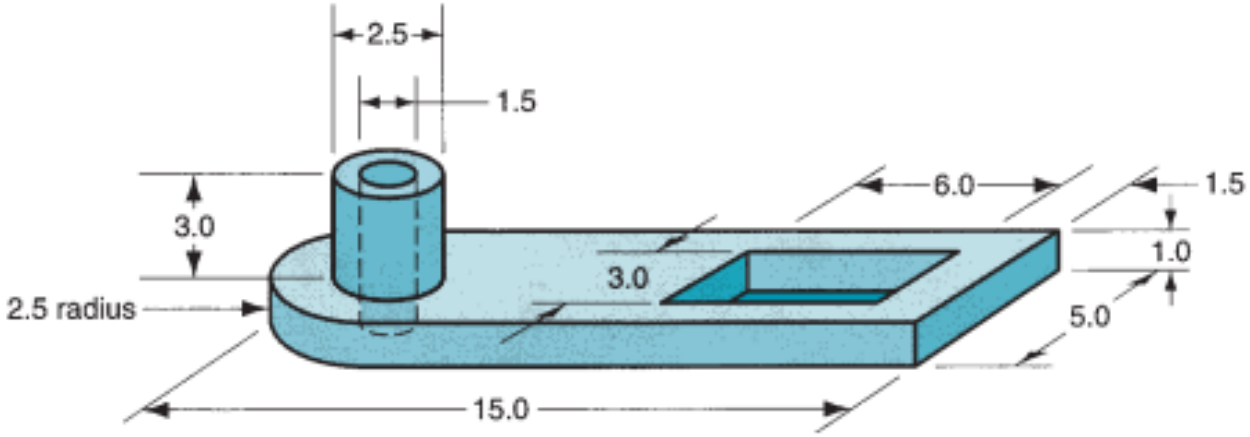
Assignment 1

The housing for a certain machinery product is made of two components, both aluminum castings. The larger component has the shape of a dish sink, and the second component is a flat cover that is attached to the first component to create an enclosed space for the machinery parts. Sand casting is used to produce the two castings, both of which are plagued by defects in the form of misruns and cold shuts. The foreman complains that the parts are too thin, and that is the reason for the defects. However, it is known that the same components are cast successfully in other foundries. What other explanation can be given for the defects?

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Assignment 2

A cylindrical riser with diameter-to-length ratio = 1.0 is to be designed for a sand casting mold. The casting geometry is as below, in which the units are inches. If the mold constant in Chvorinov's rule is $19.5\ \text{min}/in^2$, determine the dimensions of the riser so that the riser will take 0.5 min longer to freeze than the casting itself.



Thank You